

Inverse Transform Watermark

Remark 1.

Let X be a random variable with CDF $F_X(x) = \mathbb{P}(X \leq x)$.

Let $U \sim U(0, 1)$.

Let $Y = F_X^{-1}(U)$, where $F^{-1}(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}$, $\forall p \in [0, 1]$.

Then $Y \sim X$.

We see "unbiasedness" there.

How does it work?

Imposing an Order on the Vocabulary

The Inverse Transform method requires a 1D real number line to define a "Cumulative" function (i.e., you must be able to say $X \leq x$).

However, the vocabulary $\mathcal{W}$ is a set of words (categorical data) with no inherent mathematical order.

To solve this, we introduce a permutation $\pi : \mathcal{W} \rightarrow \{1, 2, \dots, |\mathcal{W}|\}$.

- $\pi(w)$ assigns a unique integer rank to a token w .
- $\pi^{-1}(i)$ takes rank i and returns the corresponding token. This permutation π allows us to arrange the discrete probabilities of the NTP distribution $\mathbf{P}$ into a 1D sequence on the integer line. The probability of the token at position i is $P_{\pi^{-1}(i)}$.

Remark 2.

Why need permutation?

Even if we use some fixed ordering, inverse transform sampling still works and the watermark is still unbiased (intuitively, only the "length" but not the position matters).

But we need this permutation to preserve security (if use some fixed order, then w_t and the hidden pseudorandom number U_t 's coupling relationship are static and clear.) and more importantly, validity of the pivotal quantity under H_0 .

Deriving the CDF $F(x; \pi)$

Consider the multinomial distribution with probability mass $P_{\pi^{-1}(i)}$ at i for $1 \leq i \leq |\mathcal{W}|$.

Now that the tokens are ordered from 1 to $|\mathcal{W}|$, we can define the CDF. The CDF $F(x; \pi)$ is the cumulative sum of probabilities for all tokens assigned a rank less than or equal to x .

For any real number x : $F(x; \pi) = \sum_{i=1}^{\lfloor x \rfloor} P_{\pi^{-1}(i)}$

To express this in terms of the original vocabulary $\mathcal{W}$ rather than the integer indices, we sum over all tokens $w' \in \mathcal{W}$, but use an indicator function $\mathbf{1}_{\{\pi(w') \leq x\}}$ to only include the probabilities of tokens whose rank is less than or equal to x .
 $F(x; \pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq x\}}$

Deriving the Generalized Inverse $F^{-1}(U; \pi)$

Because the distribution is discrete, the CDF $F(x; \pi)$ is a step function. we use the generalized inverse. For a uniform random variable $U \sim U(0, 1)$, $F^{-1}(U; \pi) = \min\{i \in \mathbb{Z} : F(i; \pi) \geq U\}$

Which is $F^{-1}(U; \pi) = \min\{i : \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq i\}} \geq U\}$

Deriving the Decoder $\mathcal{S}^{\text{inv}}$

$w_t = \mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) := \pi^{-1}(F^{-1}(U; \pi))$ where $\zeta = (\pi, U)$ and the permutation π is uniformly at random.

We show this watermark is unbiased.

We need to show $\mathbb{P}(\mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) = w) = P_w$ for all $w \in \mathcal{W}$.

Proof.

Let w be an arbitrary token in $\mathcal{W}$. Let its rank under the permutation be $k = \pi(w)$. Applying π to both sides of $\pi^{-1}(F^{-1}(U; \pi)) = \mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) = w$, equivalently $F^{-1}(U; \pi) = \pi(w) = k$.

And so equivalently $F(k-1; \pi) < U \leq F(k; \pi)$.

Since $F(k; \pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq k\}}$ and $F(k-1; \pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') < k\}}$, then equivalently

$$\sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') < k\}} < U \leq \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \leq k\}}$$

The length of this interval is $F(k; \pi) - F(k-1; \pi) = P_{\pi^{-1}(k)} = P_w$.

Since U is uniformly distributed on the interval $(0, 1)$, then $\mathbb{P}(\mathcal{S}^{\text{inv}}(\mathbf{P}, \zeta) = w) = \mathbb{P}(F(k-1; \pi) < U \leq F(k; \pi)) = P_w$.

Or we can use the first remark in this document.

$\mathbb{P}(X = i) = P_{\pi^{-1}(i)}$, if $Y = F_X^{-1}(U)$, then $Y \sim X$.

$\mathbb{P}(Y = i) = \mathbb{P}(X = i) = P_{\pi^{-1}(i)}$.

Note $\mathbb{P}(\text{output } w) = \mathbb{P}(Y = \pi(w))$, thus $\mathbb{P}(Y = \pi(w)) = \mathbb{P}(X = \pi(w)) = P_{\pi^{-1}(\pi(w))} = P_w$.

□

Main results

By "under H_1 , in contrast, a larger U_t value suggests a larger value of $\pi_t(w_t)$ in distribution for the watermarked text, which can be gleaned from (23)", it means that: recall from above, we know $\pi_t(w_t) = w$ **if and only if**

$$\sum_{w' \in \mathcal{W}} P_{t,w'} \mathbf{1}_{\{\pi_t(w') < w\}} < U_t \leq \sum_{w' \in \mathcal{W}} P_{t,w'} \mathbf{1}_{\{\pi_t(w') \leq w\}}$$

Therefore

$$\pi_t(w_t) = w \iff U_t \in \left(a_{\pi_t^{-1}, w-1}, a_{\pi_t^{-1}, w} \right]$$

Typo above equation (37). Should be $\pi(l) = w$ not $\pi(l) = k$ in the index of summation.

$$\sum_{w' \in \mathcal{W}} P_{t,w'} \mathbf{1}_{\{\pi_t(w') < w\}} < U_t \leq \sum_{w' \in \mathcal{W}} P_{t,w'} \mathbf{1}_{\{\pi_t(w') \leq w\}}$$

Proof of Lemma 4.1

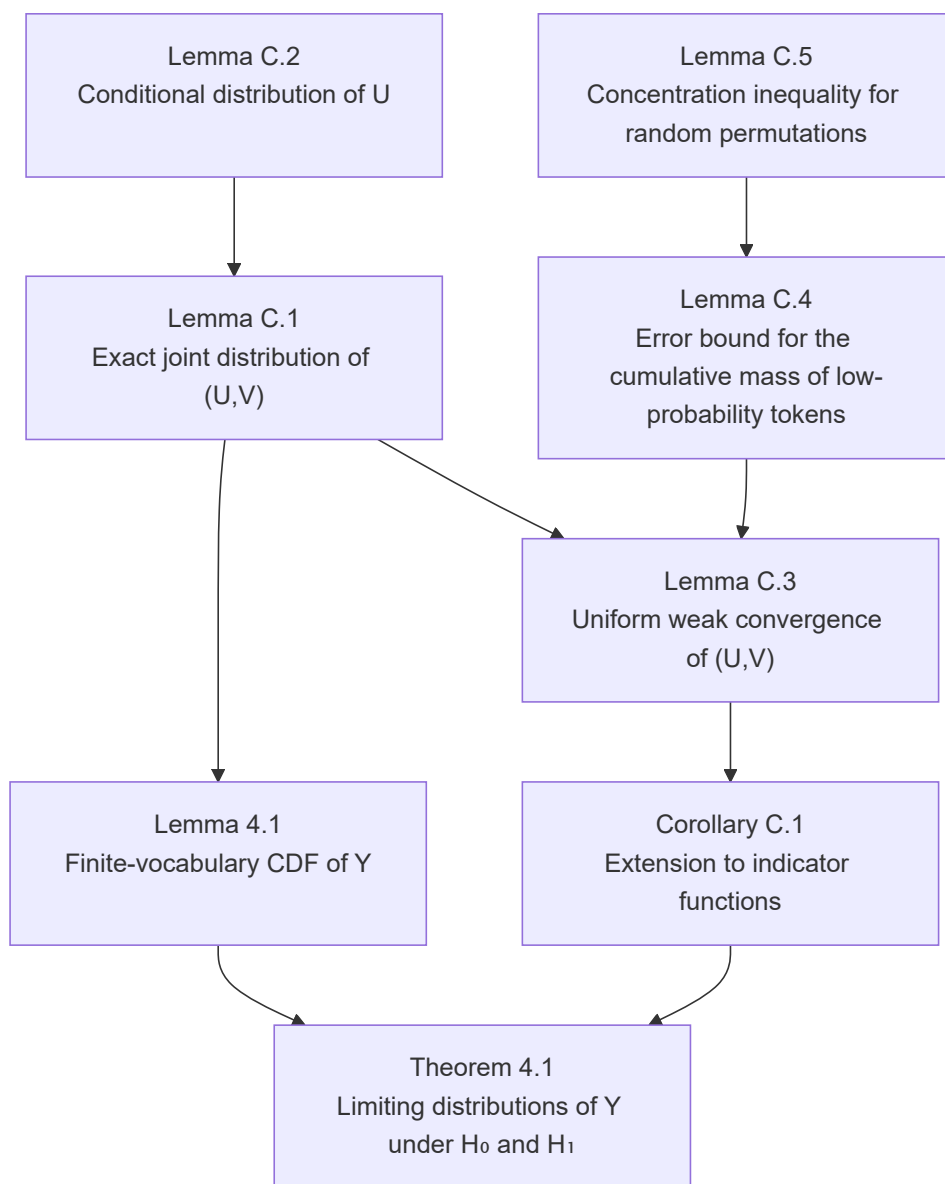
For H_0 , From

$$\mathbb{P}(Y_t^{\text{dif}} \leq r \mid H_0) = \sum_{w \in \mathcal{W}} \mathbb{P}(U_t \in [\eta(w) - r, \eta(w) + r] \cap [0, 1], \pi_t(w_t) = w \mid H_0)$$

Use results from lemma C.1 for H_0 : the joint probability is $\frac{1}{|\mathcal{W}|}$ multiplied by the probability of U_t falling into the target interval. Because $U_t \sim U(0, 1)$, the probability of U_t falling into a specific interval within $[0, 1]$ is exactly the length of that interval.

For H_1 , note changed order of summation and replaced probability to uniform measure.

Theorem 4.1



Now we first focus on Lemma C.5

Lemma C.5

First we look at Bernstein's Inequality.

Lemma 3 (Bernstein's Inequality).

Let $X_1, \dots, X_n$ be independent random variables satisfying

$$\mathbb{E}[X_i] = 0, \quad |X_i| \leq M \quad \text{almost surely}$$

for every i .

Define

$$S_n = \sum_{i=1}^n X_i, \quad V = \sum_{i=1}^n \mathbb{E}[X_i^2] = \sum_{i=1}^n \text{Var}(X_i).$$

Then, for every $t > 0$,

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + Mt/3)}\right)$$

Consequently,

$$\mathbb{P}(|S_n| \geq t) \leq 2 \exp \left(-\frac{t^2}{2(V + Mt/3)} \right)$$

Proof.

We will first show that, for every

$$0 < \lambda < \frac{3}{M}$$

we have

$$\mathbb{E}[e^{\lambda X_i}] \leq \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right) \quad (1)$$

Since X_i is bounded, by Fubini theorem, since

$$\sum_{k=0}^{\infty} \mathbb{E} \left[\left| \frac{\lambda^k X_i^k}{k!} \right| \right] \leq \sum_{k=0}^{\infty} \frac{(\lambda M)^k}{k!} = e^{\lambda M} < \infty.$$

Then

$$\mathbb{E} \left[\sum_{k=0}^{\infty} \frac{\lambda^k X_i^k}{k!} \right] = \sum_{k=0}^{\infty} \mathbb{E} \left[\frac{\lambda^k X_i^k}{k!} \right].$$

Therefore, by Taylor and above,

$$\begin{aligned} \mathbb{E}[e^{\lambda X_i}] &= \mathbb{E} \left[1 + \lambda X_i + \sum_{k=2}^{\infty} \frac{\lambda^k X_i^k}{k!} \right] \\ &= 1 + \lambda \mathbb{E}[X_i] + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}[X_i^k]}{k!}. \end{aligned}$$

Since $\mathbb{E}[X_i] = 0$,

$$\mathbb{E}[e^{\lambda X_i}] = 1 + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}[X_i^k]}{k!}.$$

For every integer $k \geq 2$,

$$\mathbb{E}[X_i^k] \leq \mathbb{E}[|X_i|^k].$$

Moreover, because $|X_i| \leq M$,

$$|X_i|^k = X_i^2 |X_i|^{k-2} \leq X_i^2 M^{k-2}$$

It follows that

$$\mathbb{E}[|X_i|^k] \leq M^{k-2} \mathbb{E}[X_i^2].$$

Thus,

$$\mathbb{E}[e^{\lambda X_i}] \leq 1 + \mathbb{E}[X_i^2] \sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{k!}$$

For every integer $k \geq 2$,

$$k! \geq 2 \cdot 3^{k-2}$$

This follows by induction. Equality holds for $k = 2$, and if it holds for k , then

$$(k+1)! = (k+1)k! \geq 3 \cdot 2 \cdot 3^{k-2} = 2 \cdot 3^{k-1}$$

Using $k! \geq 2 \cdot 3^{k-2}$, we obtain

$$\begin{aligned}\sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{k!} &\leq \sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{2 \cdot 3^{k-2}} \\ &= \frac{\lambda^2}{2} \sum_{k=2}^{\infty} \left(\frac{\lambda M}{3} \right)^{k-2}\end{aligned}$$

Since $\lambda M/3 < 1$, the geometric series converges and

$$\sum_{k=2}^{\infty} \left(\frac{\lambda M}{3} \right)^{k-2} = \frac{1}{1 - \lambda M/3}.$$

Therefore,

$$\sum_{k=2}^{\infty} \frac{\lambda^k M^{k-2}}{k!} \leq \frac{\lambda^2}{2(1 - \lambda M/3)}.$$

Substituting this into above gives

$$\mathbb{E}[e^{\lambda X_i}] \leq 1 + \frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)}.$$

Finally, using $1 + u \leq e^u$ for $u \geq 0$, we obtain

$$\mathbb{E}[e^{\lambda X_i}] \leq \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right).$$

Next, because $X_1, \dots, X_n$ are independent,

$$\begin{aligned}\mathbb{E}[e^{\lambda S_n}] &= \mathbb{E} \left[e^{\lambda \sum_{i=1}^n X_i} \right] \\ &= \mathbb{E} \left[\prod_{i=1}^n e^{\lambda X_i} \right] \\ &= \prod_{i=1}^n \mathbb{E}[e^{\lambda X_i}].\end{aligned}$$

Applying $\mathbb{E}[e^{\lambda X_i}] \leq \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right)$ to each factor gives

$$\begin{aligned}\mathbb{E}[e^{\lambda S_n}] &\leq \prod_{i=1}^n \exp \left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right) \\ &= \exp \left(\frac{\lambda^2 \sum_{i=1}^n \mathbb{E}[X_i^2]}{2(1 - \lambda M/3)} \right).\end{aligned}$$

Since

$$V = \sum_{i=1}^n \mathbb{E}[X_i^2],$$

we conclude that

$$\mathbb{E}[e^{\lambda S_n}] \leq \exp \left(\frac{\lambda^2 V}{2(1 - \lambda M/3)} \right)$$

for every $0 < \lambda < 3/M$.

Fix $t > 0$. For every $0 < \lambda < 3/M$, the monotonicity of the exponential function and Markov's inequality give

$$\begin{aligned}\mathbb{P}(S_n \geq t) &= \mathbb{P}(e^{\lambda S_n} \geq e^{\lambda t}) \\ &\leq e^{-\lambda t} \mathbb{E}[e^{\lambda S_n}].\end{aligned}$$

Using $\mathbb{E}[e^{\lambda S_n}] \leq \exp \left(\frac{\lambda^2 V}{2(1 - \lambda M/3)} \right)$,

$$\mathbb{P}(S_n \geq t) \leq \exp \left(-\lambda t + \frac{\lambda^2 V}{2(1 - \lambda M/3)} \right)$$

Assume first that $V > 0$, and choose

$$\lambda = \frac{t}{V + Mt/3}$$

This choice is admissible because

$$\frac{\lambda M}{3} = \frac{Mt/3}{V + Mt/3} < 1.$$

Let

$$D := V + \frac{Mt}{3}.$$

Then $\lambda = \frac{t}{D}$, and

$$1 - \frac{\lambda M}{3} = 1 - \frac{Mt/3}{D} = \frac{V}{D}.$$

Consequently,

$$\begin{aligned} -\lambda t + \frac{\lambda^2 V}{2(1 - \lambda M/3)} &= -\frac{t^2}{D} + \frac{(t^2/D^2)V}{2(V/D)} \\ &= -\frac{t^2}{D} + \frac{t^2}{2D} \\ &= -\frac{t^2}{2D}. \end{aligned}$$

Substituting $D = V + Mt/3$ into $\mathbb{P}(S_n \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2 V}{2(1 - \lambda M/3)}\right)$, we obtain

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + Mt/3)}\right).$$

If $V = 0$, then

$$\sum_{i=1}^n \mathbb{E}[X_i^2] = 0.$$

Since every term is nonnegative, $\mathbb{E}[X_i^2] = 0$ for every i , which implies $X_i = 0$ almost surely. Thus $S_n = 0$ almost surely, and the inequality is trivial.

This proves the one-sided Bernstein inequality.

The variables $-X_1, \dots, -X_n$ are also independent and satisfy

$$\mathbb{E}[-X_i] = 0, \quad | -X_i | \leq M, \quad \text{Var}(-X_i) = \text{Var}(X_i).$$

Applying the one-sided inequality to $-S_n$ gives

$$\mathbb{P}(S_n \leq -t) = \mathbb{P}(-S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + Mt/3)}\right).$$

Since event

$$\{|S_n| \geq t\} = \{S_n \geq t \sqcup S_n \leq -t\}$$

then

$$\begin{aligned} \mathbb{P}(|S_n| \geq t) &\leq \mathbb{P}(S_n \geq t) + \mathbb{P}(S_n \leq -t) \\ &\leq 2 \exp\left(-\frac{t^2}{2(V + Mt/3)}\right). \end{aligned}$$

This completes the proof. □

Remark 4.

We want to write the Bernstein inequality in the so called high-probability form.

$$\mathbb{P}(|S_n| \geq t) \leq 2 \exp \left(-\frac{t^2}{2(V + Mt/3)} \right) \stackrel{?}{\leq} 2e^{-x}$$

It is sufficient to require

$$\frac{t^2}{2(V + Mt/3)} \geq x$$

Since the denominator is positive,

$$t^2 \geq 2x \left(V + \frac{Mt}{3} \right)$$

Equivalently,

$$t^2 - \frac{2Mx}{3}t - 2Vx \geq 0$$

The positive root of the corresponding quadratic equation is

$$t_*(x) = \frac{Mx}{3} + \sqrt{2Vx + \frac{M^2x^2}{9}}$$

Thus, every $t \geq t_*(x)$ satisfies the inequality. This gives the exact form

$$\mathbb{P} \left(|S_n| \geq \frac{Mx}{3} + \sqrt{2Vx + \frac{M^2x^2}{9}} \right) \leq 2e^{-x}$$

Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, we have

$$\sqrt{2Vx + \frac{M^2x^2}{9}} \leq \sqrt{2Vx} + \frac{Mx}{3}$$

Therefore,

$$t_*(x) \leq \sqrt{2Vx} + \frac{2Mx}{3}$$

Choosing the slightly larger and simpler threshold

$$t = \sqrt{2Vx} + \frac{2Mx}{3}$$

gives

$$\mathbb{P} \left(|S_n| \geq \sqrt{2Vx} + \frac{2M}{3}x \right) \leq 2e^{-x}$$

Remark 5 (Lemma C.5 is the permuted version of Bernstein inequality).

Compare Bernstein to Lemma C.5,

$$\mathbb{P} \left(|S_n| \geq \sqrt{2Vx} + \frac{2M}{3}x \right) \leq 2e^{-x}$$

where $S_n = \sum_{i=1}^n X_i$.

$$\mathbb{P} \left(|Z - \mathbb{E}_\pi Z| \geq 2\sqrt{\left(\frac{1}{|\mathcal{W}|} \sum_{j,l=1}^{|\mathcal{W}|} b_{j,l}^2 \right)x} + \sup_{1 \leq j,l \leq |\mathcal{W}|} b_{j,l}x \right) \leq 8e^{1/16} \exp \left(-\frac{x}{16} \right)$$

where $Z = \sum_{j \in \mathcal{W}} b_{j,\bar{\pi}(j)}$, $\bar{\pi}$ a random uniform permutation in $\mathcal{W}$.

By random uniform permutation, it means that for every $\pi \in \Pi$,

$$\mathbb{P}(\bar{\pi} = \pi) = \frac{1}{|\mathcal{W}|!}$$

Consequently, for a fixed j , $\bar{\pi}(j)$ is uniform on $\{1, \dots, \mathcal{W}\}$, i.e.

$$\mathbb{P}(\bar{\pi}(j) = l) = \frac{1}{|\mathcal{W}|}$$

for $l = 1, \dots, |\mathcal{W}|$.

If we denote

$$V_{\text{perm}} = \frac{1}{|\mathcal{W}|} \sum_{j,l=1}^{|\mathcal{W}|} b_{j,l}^2 = \sum_{j=1}^n \mathbb{E}_{\bar{\pi}} \left[b_{j,\bar{\pi}(j)}^2 \right], \quad M_{\text{perm}} = \sup_{1 \leq j,l \leq |\mathcal{W}|} b_{j,l}$$

because (intuitively it is clear, as the permutation is uniform)

$$\begin{aligned} \mathbb{E}_{\bar{\pi}} \left[b_{j,\bar{\pi}(j)}^2 \right] &= \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \mathbb{P}(\bar{\pi}(j) = l) \\ &= \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \frac{1}{|\mathcal{W}|} \\ &= \frac{1}{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \end{aligned}$$

so

$$\begin{aligned} \sum_{j=1}^{|\mathcal{W}|} \mathbb{E}_{\bar{\pi}} \left[b_{j,\bar{\pi}(j)}^2 \right] &= \sum_{j=1}^{|\mathcal{W}|} \left(\frac{1}{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \right) \\ &= \frac{1}{|\mathcal{W}|} \sum_{j=1}^{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{W}|} b_{j,l}^2 \\ &= \frac{1}{|\mathcal{W}|} \sum_{j,l=1}^{|\mathcal{W}|} b_{j,l}^2 \end{aligned}$$

We can see Bernstein and Lemma C.5 share the same form (if neglect constants), as

$$V = \sum_{i=1}^n \mathbb{E}[X_i^2], \quad |X_i| \leq M$$

One may ask, **why not just use Bernstein instead of Lemma C.5?**

This is because for Bernstein, it is required that $X_1, \dots, X_n$ to be independent, which is not the case in Lemma C.5.

Denote $Y_j = b_{j,\bar{\pi}(j)}$. They are not independent since for example once we know $\bar{\pi}(1) = l$, then for any $j \neq 1$, we cannot have $\bar{\pi}(j) = l$.

Lemma C.4

Lemma 6 (C.4).

Define

$$I_{\pi,w} = \mathbf{1}_{\pi^{-1}(1) > w} \quad \text{and} \quad A_{\bar{\pi},w} = \sum_{j=1}^w \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$$

Then

$$\max \left\{ \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|, \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}| \right\} \leq \sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right]$$

and

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

where $P_{(i)}$ is the i -th largest probability in P and c_0 is universal positive constant.

Proof.

The purpose of Lemma C.4 is to control the approximation error.

Note it should be

$$A_{\bar{\pi},w} = \sum_{j=1}^w \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$$

instead of $A_{\bar{\pi},w} = \sum_{j \in \mathcal{W}} \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$ in the original lemma.

It is worth noting that, the proof secretly assumed (after a fixed relabeling depending only on P)

$$P_1 = P_{(1)} = 1 - \Delta$$

This is WLOG and valid because any fixed relabeling depending only on P induces a bijection on the set of permutations, under which a uniform random permutation remains uniform. Also the authors implicitly identify the vocabulary $\mathcal{W}$ with $[|\mathcal{W}|]$.

Note the definition of

$$\bar{\pi} \sim U(\Pi)$$

and $\bar{\pi}(j)$ denotes the token on the j -th entry of the permutation $\bar{\pi}$.

Thus,

$$I_{\bar{\pi},w} = \mathbf{1}_{\bar{\pi}^{-1}(1) > w} = \begin{cases} 1, & \text{if the maximum-probability token appears after position } w \text{ in the permutation } \bar{\pi} \\ 0, & \text{if the maximum-probability token appears in the first } w \text{ position in the permutation } \bar{\pi}. \end{cases}$$

and

$$A_{\bar{\pi},w} = \sum_{j=1}^w \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$$

is calculating: For a given realization of the random permutation $\bar{\pi}$, let $L = \bar{\pi}^{-1}(1)$ denote the position of the maximum-probability token, the difference between the probability mass actually accumulated up to position $\min\{w, L - 1\}$ and the amount that would have accumulated over the same positions if the remaining probability mass Δ were evenly divided among all the other tokens. Once the maximum-probability token appears, $A_{\bar{\pi},w}$ stops changing.

Now we look at what is

$$\max \left\{ \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|, \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}| \right\} \leq \sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right]$$

where $a_{\bar{\pi},w} = \sum_{j=1}^w P_{\bar{\pi}(j)}$.

We see that

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

is the expectation, with respect to the uniformly random permutation $\bar{\pi}$, of the maximum absolute difference, taken over all positions before the maximum-probability token appears, between the probability accumulated up to position w and what would have accumulated if the remaining probability mass Δ were evenly divided among all the other tokens.

Also note

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}|$$

To see it, let

$$L = \bar{\pi}^{-1}(1), \quad D_w = a_{\bar{\pi},w} - \frac{w\Delta}{N-1}.$$

Since $L = \bar{\pi}^{-1}(1)$ is a function of the random permutation $\bar{\pi}$, L **is itself a random variable**. Consequently, taking the expectation $\mathbb{E}_{\bar{\pi}}$ averages over the randomness of L **as well as** the remaining randomness in $\bar{\pi}$.

Then

$$I_{\bar{\pi},w} |D_w| = \begin{cases} |D_w|, & w < L, \\ 0, & w \geq L. \end{cases}$$

Thus

$$\sup_w I_{\bar{\pi},w} |D_w| = \sup_{w < L} |D_w|.$$

Whereas

$$A_{\bar{\pi},w} = \begin{cases} D_w, & w < L, \\ D_{L-1}, & w \geq L. \end{cases}$$

Thus

$$\sup_w |A_{\bar{\pi},w}| = \max \left\{ \sup_{w < L} |D_w|, |D_{L-1}| \right\} = \sup_{w < L} |D_w|.$$

Therefore

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} |A_{\bar{\pi},w}|$$

as desired.

Note

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \mathbb{E}_{\bar{\pi}} \sup_{w \leq L-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

Now we look at what is

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right]$$

The difference is that we fix the maximum-probability token at each position ℓ , and we take the sup over all ℓ in the end. Intuitively, we are taking the worst case over all possible positions ℓ . This is also why intuitively the inequality holds, since the LHS is averaging all positions.

Specifically:

Recall

$$\sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| = \sup_{w \leq L-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|.$$

Since $\bar{\pi} \sim U(\Pi)$, then

$$\mathbb{P}_{\bar{\pi}}(\bar{\pi}(\ell) = 1) = \frac{1}{|\mathcal{W}|}.$$

and so using tower rule,

$$\begin{aligned}
& \mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi}, w} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \\
&= \mathbb{E}_{\bar{\pi}} \left[\mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi}, w} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| L \right] \right] \\
&= \sum_{\ell \in \mathcal{W}} \mathbb{P}_{\bar{\pi}}(L = \ell) \mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi}, w} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| L = \ell \right] \\
&= \sum_{\ell \in \mathcal{W}} \mathbb{P}_{\bar{\pi}}(\bar{\pi}(\ell) = 1) \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \\
&= \frac{1}{|\mathcal{W}|} \sum_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \\
&\leq \sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right].
\end{aligned}$$

The last inequality is due to

$$\frac{1}{|\mathcal{W}|} \sum_{\ell \in \mathcal{W}} x_{\ell} \leq \sup_{\ell \in \mathcal{W}} x_{\ell}$$

Finally we look at

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

We note that

$$\mathbb{E}_{\bar{\pi}} [a_{\bar{\pi}, w} | \bar{\pi}(\ell) = 1] = \frac{w\Delta}{|\mathcal{W}| - 1}$$

This is because:

Fix $w \leq \ell - 1$, for any $j \leq w$ and any $i \neq 1$, we have

$$\begin{aligned}
& \mathbb{P}_{\bar{\pi}} (\bar{\pi}(j) = i | \bar{\pi}(\ell) = 1) \\
&= \frac{\#\{\bar{\pi} : \bar{\pi}(\ell) = 1, \bar{\pi}(j) = i\}}{\#\{\bar{\pi} : \bar{\pi}(\ell) = 1\}} \\
&= \frac{(|\mathcal{W}| - 2)!}{(|\mathcal{W}| - 1)!} \\
&= \frac{1}{|\mathcal{W}| - 1}
\end{aligned}$$

Then

$$\begin{aligned}
\mathbb{E}_{\pi} [P_{\bar{\pi}(j)} | \bar{\pi}(\ell) = 1] &= \sum_{i=2}^{|\mathcal{W}|} P_i \mathbb{P}_{\pi} (\bar{\pi}(j) = i | \bar{\pi}(\ell) = 1) \\
&= \frac{1}{|\mathcal{W}| - 1} \sum_{i=2}^{|\mathcal{W}|} P_i \\
&= \frac{\Delta}{|\mathcal{W}| - 1}
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathbb{E}_{\bar{\pi}} [a_{\bar{\pi},w} | \bar{\pi}(\ell) = 1] &= \mathbb{E}_{\bar{\pi}} \left[\sum_{j=1}^w P_{\bar{\pi}(j)} \mid \bar{\pi}(\ell) = 1 \right] \\
&= \sum_{j=1}^w \mathbb{E}_{\bar{\pi}} [P_{\bar{\pi}(j)} | \bar{\pi}(\ell) = 1] \\
&= \sum_{j=1}^w \frac{\Delta}{|\mathcal{W}| - 1} \\
&= \frac{w\Delta}{|\mathcal{W}| - 1}
\end{aligned}$$

and we are done.

We observe that, conditional on $\bar{\pi}(\ell) = 1$, for each fixed $w \leq \ell - 1$,

$$|a_{\bar{\pi},w} - \mathbb{E}_{\bar{\pi}} [a_{\bar{\pi},w} | \bar{\pi}(\ell) = 1]| = \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

This form of "sum - its expectation" suggests that we need to use some sort of concentration inequality.

Now we need to find the "right" inequality to use.

We observe

$$0 \leq P_{\bar{\pi}(j)} \leq P_{(2)}$$

and

$$\frac{\Delta}{|\mathcal{W}| - 1} = \frac{1}{|\mathcal{W}| - 1} \sum_{i=2}^{|\mathcal{W}|} P_i$$

which implies

$$0 \leq \frac{\Delta}{|\mathcal{W}| - 1} \leq P_{(2)}$$

Combining above,

$$P_{\bar{\pi}(j)}, \frac{\Delta}{|\mathcal{W}| - 1} \in [0, P_{(2)}].$$

Since the difference between any two intergers in the interval $[0, P_{(2)}]$ is at most $P_{(2)}$,

$$\left| P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right| \leq P_{(2)}$$

This reminds us the bounded-summand in the Bernstein inequality (**One can also think of Hoeffding, but why the paper did not use Hoeffding?**).

But this raises another problem, $P_{\bar{\pi}(j)}$ is not independent.

The paper found Lemma C.5 to solve this problem.

To prove the last inequality in Lemma C.4:

As in the paper, let

$$b_{j,l} = P_l \mathbf{1}_{j \leq w, l \neq j}$$

Condition on $\bar{\pi}(\ell) = 1$ and under $w \leq \ell - 1$,

$$\begin{aligned}
Z &:= \sum_{j \in \mathcal{W}} b_{j, \bar{\pi}(j)} = \sum_{j \in \mathcal{W}} P_{\bar{\pi}(j)} \mathbf{1}_{j \leq w, \bar{\pi}(j) \neq 1} \\
&= \sum_{j=1}^w P_{\bar{\pi}(j)} \\
&= a_{\bar{\pi},w}
\end{aligned}$$

If $b_{j,l} \neq 0$, we must have $l \neq 1$. In this case, $P_l \leq P_{(2)}$, so

$$\sup_{j,l} b_{j,l} \leq P_{(2)}$$

We have

$$\begin{aligned} \sum_{j,l} b_{j,l}^2 &= \sum_{j,l} P_l^2 \mathbf{1}_{j \leq w, l \neq 1} \\ &= \sum_{j=1}^w \sum_{l \neq 1} P_l^2 \\ &= w \sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \end{aligned}$$

Condition on $\bar{\pi}(\ell) = 1$, apply Lemma C.5 to the remaining $|\mathcal{W}| - 1$ uniform permutation (because after conditioning on $\bar{\pi}(\ell) = 1$, $\bar{\pi}$ is no longer an uniform permutation on $\mathcal{W}$, so in this case cannot use Lemma C.5). Thus for any $x > 0$,

$$\mathbb{P}_{\bar{\pi}} \left(\left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq 2 \sqrt{\frac{w}{|\mathcal{W}| - 1} \left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) x + P_{(2)} x} \middle| \bar{\pi}(\ell) = 1 \right) \leq 8e^{1/16} \exp \left(-\frac{x}{16} \right)$$

Given $0 < \delta < 1$, choose

$$x = 16 \log \left(\frac{8e^{1/16}}{\delta} \right)$$

And so

$$\delta = 8e^{1/16} \exp \left(-\frac{x}{16} \right)$$

in the above inequality.

Since

$$\frac{1}{|\mathcal{W}| - 1} \leq \frac{2}{|\mathcal{W}|}$$

We get

$$\mathbb{P}_{\bar{\pi}} \left(\left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\frac{w}{|\mathcal{W}|} \left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log \left(\frac{c_0}{\delta} \right) + P_{(2)} c_0 \log \left(\frac{c_0}{\delta} \right)} \middle| \bar{\pi}(\ell) = 1 \right) \leq \delta$$

for a suitable c_0 .

Since

$$\frac{w}{|\mathcal{W}|} \leq 1$$

We get

$$\mathbb{P}_{\bar{\pi}} \left(\left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log \left(\frac{c_0}{\delta} \right) + P_{(2)} c_0 \log \left(\frac{c_0}{\delta} \right)} \middle| \bar{\pi}(\ell) = 1 \right) \leq \delta$$

Note

$$\left\{ \sup_{w \leq \ell-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq B \right\} = \bigcup_{w \leq \ell-1} \left\{ \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq B \right\},$$

where $B = \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log \left(\frac{c_0}{\delta} \right) + P_{(2)} c_0 \log \left(\frac{c_0}{\delta} \right)}$.

Note for a fixed ℓ , there exists at most $|\mathcal{W}|$ of w 's such that $w \leq \ell - 1$.

Since $\mathbb{P}(\bigcup_{i=1}^m E_i) \leq \sum_{i=1}^m \mathbb{P}(E_i)$ for event $E_1, \dots, E_m$,

$$\begin{aligned} & \mathbb{P}_{\bar{\pi}} \left(\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log\left(\frac{c_0}{\delta}\right) + P_{(2)} c_0 \log\left(\frac{c_0}{\delta}\right)} \middle| \bar{\pi}(\ell) = 1 \right) \\ & \leq \sum_{w \leq \ell-1} \delta \\ & = (\ell - 1)\delta \\ & \leq (|\mathcal{W}| - 1)\delta \\ & \leq |\mathcal{W}|\delta \end{aligned}$$

Now we choose

$$\delta = \frac{1}{|\mathcal{W}|^2}$$

then

$$|\mathcal{W}|\delta = \frac{1}{|\mathcal{W}|}$$

and

$$\log \frac{c_0}{\delta} = \log(c_0 |\mathcal{W}|^2)$$

We can write $c_0 \log(c_0 |\mathcal{W}|^2)$ as

$$c_0 \log(c_0 |\mathcal{W}|)$$

because c_0 is just a universal positive constant.

Therefore

$$\mathbb{P}_{\pi} \left(\sup_{w \leq \ell-1} \left| a_{\pi, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \geq \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)} \middle| \bar{\pi}(\ell) = 1 \right) \leq \frac{1}{|\mathcal{W}|}$$

Note that condition on $\bar{\pi}(\ell) = 1$, for all $w \leq \ell - 1$,

$$0 \leq a_{\pi, w} \leq \Delta$$

and

$$0 \leq \frac{w\Delta}{|\mathcal{W}| - 1} \leq \Delta$$

Thus

$$0 \leq \sup_{w \leq \ell-1} \left| a_{\pi, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \leq \Delta \leq 1$$

which follows from a similar argument above.

Denote

$$X_{\ell} := \sup_{w \leq \ell-1} \left| a_{\pi, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right|$$

and

$$B := \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

Define

$$\mathcal{G}_{\ell} := \{X_{\ell} < B\}$$

and

$$\mathcal{G}_\ell^c = \{X_\ell \geq B\}$$

Then from before we have

$$\mathbb{P}_\pi(\mathcal{G}_\ell^c | \bar{\pi}(\ell) = 1) \leq \frac{1}{|\mathcal{W}|}$$

Note

$$\mathbb{E}_{\bar{\pi}}[X_\ell | \bar{\pi}(\ell) = 1] = \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell} | \bar{\pi}(\ell) = 1] + \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell^c} | \bar{\pi}(\ell) = 1]$$

On $\mathcal{G}_\ell$, $X_\ell < B$, so

$$X_\ell \mathbf{1}_{\mathcal{G}_\ell} \leq B \mathbf{1}_{\mathcal{G}_\ell}$$

Thus

$$\begin{aligned} \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell} | \bar{\pi}(\ell) = 1] &\leq B \mathbb{E}_{\bar{\pi}}[\mathbf{1}_{\mathcal{G}_\ell} | \bar{\pi}(\ell) = 1] \\ &= B \mathbb{P}_\pi(\mathcal{G}_\ell | \bar{\pi}(\ell) = 1) \\ &\leq B \end{aligned}$$

On $\mathcal{G}_\ell^c$, recall from above it is always true that $X_\ell \leq 1$, and so

$$X_\ell \mathbf{1}_{\mathcal{G}_\ell^c} \leq \mathbf{1}_{\mathcal{G}_\ell^c}$$

Thus

$$\begin{aligned} \mathbb{E}_{\bar{\pi}}[X_\ell \mathbf{1}_{\mathcal{G}_\ell^c} | \bar{\pi}(\ell) = 1] &\leq \mathbb{E}_{\bar{\pi}}[\mathbf{1}_{\mathcal{G}_\ell^c} | \bar{\pi}(\ell) = 1] \\ &= \mathbb{P}_\pi(\mathcal{G}_\ell^c | \bar{\pi}(\ell) = 1) \\ &\leq \frac{1}{|\mathcal{W}|} \end{aligned}$$

Combining above, we have

$$\mathbb{E}_{\bar{\pi}}[X_\ell | \bar{\pi}(\ell) = 1] \leq B + \frac{1}{|\mathcal{W}|}$$

which is exactly

$$\mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

The intuition here is that

$$0 \leq X \leq 1, \quad \mathbb{P}(X \geq B) \leq \varepsilon \quad \implies \quad \mathbb{E}[X] \leq B + \varepsilon$$

Since this holds for all fixed ℓ , and the RHS is free of ℓ , we have

$$\sup_{\ell \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\sup_{w \leq \ell-1} \left| a_{\bar{\pi}, w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \middle| \bar{\pi}(\ell) = 1 \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

And we are finally done. □

Lemma C.3

Lemma 7 (C.3).

Let $\mathcal{J}$ collect any 1-Lipschitz-continuous function on $[0, 1]^2$. For any $J \in \mathcal{J}$, $J(U_t, \eta(\pi_t(w_t)))$ is a bounded random variable, where $\eta(w) = \frac{w-1}{|\mathcal{W}|-1}$.

Let P_t be the NTP distribution that w_t follows. We introduce a new belief class, denoted by $\mathcal{Q}_\Delta$, with the following definition:

$$\mathcal{Q}_\Delta = \{P : P_{(1)} = 1 - \Delta, \log |\mathcal{W}| \cdot P_{(2)} \leq \varepsilon_{|\mathcal{W}|}\}$$

The following equation holds uniformly for $P_t \in \mathcal{Q}_\Delta$ and $\Delta \in \left[0, 1 - \frac{1}{|\mathcal{W}|}\right]$:

$$\begin{aligned} \lim_{|\mathcal{W}| \rightarrow \infty} \mathbb{E}_{1, P_t} J(U_t, \eta(\pi_t(w_t))) &= \Delta \int_0^1 (1-x) J(\Delta x, x) dx \\ &+ \Delta \int_0^1 x J(\Delta x + 1 - \Delta, x) dx \\ &+ \int_0^1 dy \int_{\Delta y}^{\Delta y + 1 - \Delta} J(x, y) dx \end{aligned} \quad (39)$$

The uniform convergence holds in the sense that

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{J \in \mathcal{J}} \sup_{\Delta \in [0, 1 - \frac{1}{|\mathcal{W}|}]} \sup_{P_t \in \mathcal{Q}_\Delta} |\mathbb{E}_{1, P_t} J(U_t, \eta(\pi_t(w_t))) - \text{right-hand side of (39)}| = 0$$

Proof.

WLOG, relabel the tokens so that $P_1 = P_{(1)} = 1 - \Delta$.

A continuous function on a compact set is bounded. This is why "For any $J \in \mathcal{J}$, we know that J is 1-Lipschitz-continuous and bounded".

How do we get

$$\mathbb{E}_{1, P_t} J(U_t, \eta(\pi_t(w_t))) = \frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\pi \in \Pi} \int_{a_{\pi, w-1}}^{a_{\pi, w}} J(r, \eta(w)) dr \quad (1)$$

This is because we note that

$$\mathbb{E}_{1, P_t} [J(U_t, \eta(\pi_t(w_t))) \mid \pi_t = \pi] = \sum_{w \in \mathcal{W}} \mathbb{E}_{1, P_t} [J(U_t, \eta(w)) \mathbf{1}_{\pi_t(w_t)=w} \mid \pi_t = \pi]$$

So it suffices to only consider

$$\mathbb{E}_{1, P_t} [J(U_t, \eta(w)) \mathbf{1}_{\pi_t(w_t)=w} \mid \pi_t = \pi]$$

Recall

$$\pi_t(w_t) = w \iff U_t \in \left(a_{\pi_t^{-1}, w-1}, a_{\pi_t^{-1}, w}\right]$$

So

$$\begin{aligned} \mathbb{E}_{1, P_t} [J(U_t, \eta(w)) \mathbf{1}_{\pi_t(w_t)=w}] &= \mathbb{E}_{1, P_t} \left[J(U_t, \eta(w)) \mathbf{1}_{U_t \in \left(a_{\pi_t^{-1}, w-1}, a_{\pi_t^{-1}, w}\right]} \right] \\ &= \int_0^1 J(r, \eta(w)) \mathbf{1}_{\left\{a_{\pi_t^{-1}, w-1} < r \leq a_{\pi_t^{-1}, w}\right\}} \cdot 1 dr \\ &= \int_{a_{\pi_t^{-1}, w-1}}^{a_{\pi_t^{-1}, w}} J(r, \eta(w)) dr \end{aligned}$$

Thus we get

$$\mathbb{E}_{1, P_t} [J(U_t, \eta(\pi_t(w_t))) \mid \pi_t = \pi] = \sum_{w \in \mathcal{W}} \int_{a_{\pi_t^{-1}, w-1}}^{a_{\pi_t^{-1}, w}} J(r, \eta(w)) dr$$

Finally since by tower rule, we get

$$\begin{aligned}
\mathbb{E}_{1,P_t}[J(U_t, \eta(\pi_t(w_t)))] &= \sum_{\pi \in \Pi} \mathbb{P}(\pi_t = \pi) \mathbb{E}_{1,P_t}[J(U_t, \eta(\pi_t(w_t))) \mid \pi_t = \pi] \\
&= \sum_{\pi \in \Pi} \mathbb{P}(\pi_t = \pi) \sum_{w \in \mathcal{W}} \int_{a_{\pi_t, w-1}^{\pi_t-1, w}}^{a_{\pi_t-1, w}} J(r, \eta(w)) dr \\
&= \sum_{\pi \in \Pi} \mathbb{P}(\pi_t = \pi) \sum_{w \in \mathcal{W}} \int_{a_{\pi_t, w-1}}^{a_{\pi_t, w}} J(r, \eta(w)) dr \\
&= \frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\pi \in \Pi} \int_{a_{\pi, w-1}}^{a_{\pi, w}} J(r, \eta(w)) dr
\end{aligned}$$

One can also use Lemma C.1. Recall

$$\mathbb{P}_1(U_t \leq r, \pi_t(w_t) = w) = \frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \lambda((a_{\pi, w-1}, a_{\pi, w}] \cap [0, r])$$

where λ is the Lebesgue measure.

For fixed $w \in \mathcal{W}$, define

$$\mu_w(B) := \mathbb{P}_1(U_t \in B, \pi_t(w_t) = w), \quad B \subseteq [0, 1]$$

so we see that Lemma C.1. is saying (note $\mu_w \ll \lambda$),

$$\begin{aligned}
\mu_w(B) &= \mathbb{P}_1(U_t \in B, \pi_t(w_t) = w) = \frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \lambda((a_{\pi, w-1}, a_{\pi, w}] \cap B) \\
&= \frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \int_B \mathbf{1}_{a_{\pi, w-1} < r \leq a_{\pi, w}} dr \\
&= \int_B \left[\frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \mathbf{1}_{a_{\pi, w-1} < r \leq a_{\pi, w}} \right] dr
\end{aligned}$$

Thus

$$g_w(r) := \frac{d\mu_w}{dr}(r) = \frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \mathbf{1}_{a_{\pi, w-1} < r \leq a_{\pi, w}}$$

We can view

$$g(r, w) := g_w(r)$$

Therefore we obtained the joint pdf

$$f_{U_t, \pi_t(w_t)}(r, w) = \frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \mathbf{1}_{a_{\pi, w-1} < r \leq a_{\pi, w}}$$

Thus it easily follows that

$$\begin{aligned}
\mathbb{E}_{1,P_t}[J(U_t, \eta(\pi_t(w_t)))] &= \sum_{w \in \mathcal{W}} \int_0^1 J(r, \eta(w)) f_{U_t, \pi_t(w_t)}(r, w) dr \\
&= \sum_{w \in \mathcal{W}} \int_0^1 J(r, \eta(w)) \left[\frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \mathbf{1}_{a_{\pi, w-1} < r \leq a_{\pi, w}} \right] dr \\
&= \frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\pi \in \Pi} \int_{a_{\pi, w-1}}^{a_{\pi, w}} J(r, \eta(w)) dr
\end{aligned}$$

Therefore we finally have shown (1).

Why do we split $\mathbb{E}_{1,P_t}[J(U_t, \eta(\pi_t(w_t)))]$ into 3 cases, $\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3$?

The length of the w -th interval is

$$a_{\pi, w} - a_{\pi, w-1} = P_{\pi(w)}$$

If $w \neq l$, then

$$P_{\pi(w)} \leq P_{(2)}$$

In the assumption of Theorem 4.1, under Q_Δ , $P_{(2)}$ goes to 0 as $|\mathcal{W}|$ goes to ∞ .

But if $w = l$, then

$$a_{\pi,w} - a_{\pi,w-1} = P_1 = 1 - \Delta$$

which could be $O(1)$ and won't disappear as $|\mathcal{W}| \rightarrow \infty$. This is a "large jump".

Consider $\mathcal{T}_1$ (the case where $l > w$, note there is a typo in the original proof).

The Taylor expansion together with the boundedness of J implies that

$$\int_{a_{\pi,w-1}}^{a_{\pi,w}} J(r, \eta(w)) dr = J(a_{\pi,w}, \eta(w)) P_{\pi(w)} + O(P_{\pi(w)}^2) \quad (2)$$

Why?

We can think of

$$\int_{a_{\pi,w-1}}^{a_{\pi,w}} J(r, \eta(w)) dr \approx J(a_{\pi,w}, \eta(w)) P_{\pi(w)}$$

That is, intuitively,

$$J(r, \eta(w)) = \underbrace{J(a_{\pi,w}, \eta(w))}_{\text{Approximation on the interval}} + \underbrace{[J(r, \eta(w)) - J(a_{\pi,w}, \eta(w))]}_{\text{Change on the interval}}$$

And

$$\begin{aligned} \int_{a_{\pi,w-1}}^{a_{\pi,w}} J(a_{\pi,w}, \eta(w)) dr &= J(a_{\pi,w}, \eta(w)) \cdot (a_{\pi,w} - a_{\pi,w-1}) \\ &= J(a_{\pi,w}, \eta(w)) P_{\pi(w)} \end{aligned}$$

By 1-Lipschitz,

$$\begin{aligned} \left| \int_{a_{\pi,w-1}}^{a_{\pi,w}} J(r, \eta(w)) dr - J(a_{\pi,w}, \eta(w)) P_{\pi(w)} \right| &= \left| \int_{a_{\pi,w-1}}^{a_{\pi,w}} [J(r, \eta(w)) - J(a_{\pi,w}, \eta(w))] dr \right| \\ &\leq \int_{a_{\pi,w-1}}^{a_{\pi,w}} |J(r, \eta(w)) - J(a_{\pi,w}, \eta(w))| dr \\ &\leq \int_{a_{\pi,w-1}}^{a_{\pi,w}} |r - a_{\pi,w}| dr \\ &= \int_{a_{\pi,w-1}}^{a_{\pi,w}} (a_{\pi,w} - r) dr \\ &= \frac{1}{2} P_{\pi(w)}^2 \end{aligned}$$

By definition of Big O , we have shown (2).

Next, we need to show

$$\frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} O(P_{\pi(w)}^2) = O(P_{(2)}) \quad (3)$$

Define

$$R_{\pi,w} := \int_{a_{\pi,w-1}}^{a_{\pi,w}} J(r, \eta(w)) dr - J(a_{\pi,w}, \eta(w)) P_{\pi(w)}$$

For fixed π , if $\pi^{-1}(1) > w$, then $\pi(w) \neq 1$. Then

$$\sum_{w < \pi^{-1}(1)} P_{\pi(w)}^2 \leq \sum_{j \neq 1} P_j^2$$

Since for all $j \neq 1$, $P_j \leq P_{(2)}$, then

$$P_j^2 \leq P_{(2)} P_j$$

Thus

$$\begin{aligned} \sum_{j \neq 1} P_j^2 &\leq P_{(2)} \sum_{j \neq 1} P_j \\ &= P_{(2)} \Delta \\ &\leq P_{(2)} \end{aligned}$$

That is, for a fixed π ,

$$\sum_{w < \pi^{-1}(1)} P_{\pi(w)}^2 \leq P_{(2)}$$

Now we show (3).

Note from (2), we know

$$|R_{\pi,w}| \leq C P_{\pi(w)}^2$$

So

$$\left| \frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} R_{\pi,w} \right| \leq \frac{C}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} P_{\pi(w)}^2$$

For a fixed w ,

$$\sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} P_{\pi(w)}^2 = \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi} P_{\pi(w)}^2 \mathbf{1}_{\pi \in \Pi_\ell}$$

It is valid that

$$\sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi} P_{\pi(w)}^2 \mathbf{1}_{\pi \in \Pi_\ell} = \sum_{\pi \in \Pi} P_{\pi(w)}^2 \sum_{\ell=w+1}^{|\mathcal{W}|} \mathbf{1}_{\pi \in \Pi_\ell}$$

And

$$\sum_{\ell=w+1}^{|\mathcal{W}|} \mathbf{1}_{\pi \in \Pi_\ell} = \sum_{\ell=w+1}^{|\mathcal{W}|} \mathbf{1}_{\pi(\ell)=1} = \mathbf{1}_{\pi^{-1}(1) > w}$$

because:

- for each π , by definition exists a unique $\ell = \pi^{-1}(1)$.
- If $\pi^{-1}(1) > w$, then the only ℓ such that $\pi(\ell) = 1$ would be in $\{w+1, \dots, |\mathcal{W}|\}$.
- If $\pi^{-1}(1) \leq w$, then no ℓ in $\{w+1, \dots, |\mathcal{W}|\}$ would satisfy $\pi(\ell) = 1$.

Therefore combine above we get

$$\sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} P_{\pi(w)}^2 = \sum_{\pi \in \Pi} P_{\pi(w)}^2 \mathbf{1}_{\pi^{-1}(1) > w}$$

Thus

$$\sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} P_{\pi(w)}^2 = \sum_{w \in \mathcal{W}} \sum_{\pi \in \Pi} P_{\pi(w)}^2 \mathbf{1}_{\pi^{-1}(1) > w}$$

Then

$$\sum_{w \in \mathcal{W}} \sum_{\pi \in \Pi} P_{\pi(w)}^2 \mathbf{1}_{\pi^{-1}(1) > w} = \sum_{\pi \in \Pi} \sum_{w \in \mathcal{W}} P_{\pi(w)}^2 \mathbf{1}_{w < \pi^{-1}(1)} = \sum_{\pi \in \Pi} \sum_{w < \pi^{-1}(1)} P_{\pi(w)}^2$$

Therefore we get

$$\frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} P_{\pi(w)}^2 = \frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \sum_{w < \pi^{-1}(1)} P_{\pi(w)}^2$$

So we showed

$$\begin{aligned} \left| \frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} R_{\pi,w} \right| &\leq \frac{C}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \sum_{w < \pi^{-1}(1)} P_{\pi(w)}^2 \\ &\leq \frac{C}{|\mathcal{W}|!} \sum_{\pi \in \Pi} P_{(2)} \\ &= \frac{C}{|\mathcal{W}|!} |\mathcal{W}|! P_{(2)} \\ &= CP_{(2)} \end{aligned}$$

By definition of Big O ,

$$\frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} R_{\pi,w} = O(P_{(2)})$$

Recall

$$R_{\pi,w} = O(P_{\pi(w)}^2)$$

Therefore

$$\frac{1}{|\mathcal{W}|!} \sum_{w \in \mathcal{W}} \sum_{\ell=w+1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_\ell} O(P_{\pi(w)}^2) = O(P_{(2)})$$

And so we have prove (3).

Now we show (41).

$$\begin{aligned} &\left| \sum_{w \in \mathcal{W}} \left[\underbrace{J(a_{\bar{\pi},w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi},w}}_{\text{real term}} - \underbrace{J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w)\right) \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1}}_{\text{"average" approximation}} \right] \right| \\ &\leq \left| \sum_{w \in \mathcal{W}} J(a_{\bar{\pi},w}, \eta(w)) \left(P_{\bar{\pi}(w)} - \frac{\Delta}{|\mathcal{W}|-1} \right) I_{\bar{\pi},w} \right| + O(1) \sum_{w \in \mathcal{W}} \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \end{aligned} \quad (41)$$

Consider

$$\begin{aligned} &J(a_{\bar{\pi},w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi},w} - J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w)\right) \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \\ &= J(a_{\bar{\pi},w}, \eta(w)) \left(P_{\bar{\pi}(w)} - \frac{\Delta}{|\mathcal{W}|-1} \right) I_{\bar{\pi},w} + \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \left[J(a_{\bar{\pi},w}, \eta(w)) - J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w)\right) \right] \end{aligned}$$

Then

$$\begin{aligned} &\left| \sum_{w \in \mathcal{W}} \left[J(a_{\bar{\pi},w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi},w} - J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w)\right) \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \right] \right| \\ &\leq \left| \sum_{w \in \mathcal{W}} J(a_{\bar{\pi},w}, \eta(w)) \left(P_{\bar{\pi}(w)} - \frac{\Delta}{|\mathcal{W}|-1} \right) I_{\bar{\pi},w} \right| + \sum_{w \in \mathcal{W}} \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \left| J(a_{\bar{\pi},w}, \eta(w)) - J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w)\right) \right| \\ &\leq \underbrace{\left| \sum_{w \in \mathcal{W}} J(a_{\bar{\pi},w}, \eta(w)) \left(P_{\bar{\pi}(w)} - \frac{\Delta}{|\mathcal{W}|-1} \right) I_{\bar{\pi},w} \right|}_{\text{single-step probability deviation}} + \underbrace{\sum_{w \in \mathcal{W}} \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right|}_{\text{cumulative endpoint deviation}} \end{aligned}$$

which is (41).

The proof in the paper includes an extra $O(1)$. Maybe the paper accidentally assumed J to be L -Lipschitz, but should be 1-Lipschitz by assumption.

Correction to a typo: should be

$$A_{\bar{\pi},w} = \sum_{j=1}^w \left(P_{\bar{\pi}(j)} - \frac{\Delta}{|\mathcal{W}| - 1} \right) I_{\bar{\pi},j}$$

What does " $A_{\bar{\pi},w}$ viewed as a centered version of $a_{\bar{\pi},w}$ " mean?

Well, for $\mathcal{T}_1$ where $I_{\bar{\pi},w} = 1$, directly

$$\underbrace{A_{\bar{\pi},w}}_{\text{centered fluctuation}} = a_{\bar{\pi},w} - \underbrace{\frac{w\Delta}{|\mathcal{W}| - 1}}_{\mathbb{E}_{\bar{\pi}}[a_{\bar{\pi},w} | \bar{\pi}(\ell)=1]}$$

Note that

$$\mathbb{P}_{\bar{\pi}}(\bar{\pi}(w) = j \mid I_{\bar{\pi},w} = 1) = \frac{1}{|\mathcal{W}| - 1}, \quad j \neq 1$$

and so

$$\begin{aligned} \mathbb{E}_{\bar{\pi}}[P_{\bar{\pi}(w)} \mid I_{\bar{\pi},w} = 1] &= \sum_{j \neq 1} P_j \mathbb{P}_{\bar{\pi}}(\bar{\pi}(w) = j \mid I_{\bar{\pi},w} = 1) \\ &= \sum_{j \neq 1} P_j \frac{1}{|\mathcal{W}| - 1} \\ &= \frac{\sum_{j \neq 1} P_j}{|\mathcal{W}| - 1} \\ &= \frac{\Delta}{|\mathcal{W}| - 1} \end{aligned}$$

then it follows that

$$\mathbb{E}_{\bar{\pi}}[a_{\bar{\pi},w} \mid I_{\bar{\pi},w} = 1] = \frac{w\Delta}{|\mathcal{W}| - 1}$$

which is important for intuition.

It seems that we have not make use of $P_{(2)}$.

Well, we will see that

$$\mathbb{E}_{\bar{\pi}} \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}| - 1} \right| \rightarrow 0$$

which is a much stronger condition.

Note

$$\begin{aligned} \sum_{w=1}^{|\mathcal{W}|} B_w (A_w - A_{w-1}) &= \sum_{w=1}^{|\mathcal{W}|} B_w A_w - \sum_{w=1}^{|\mathcal{W}|} B_w A_{w-1} \\ &= \sum_{w=1}^{|\mathcal{W}|} B_w A_w - \sum_{w=0}^{|\mathcal{W}|-1} B_{w+1} A_w \\ &= \sum_{w=1}^{|\mathcal{W}|-1} (B_w - B_{w+1}) A_w + B_{|\mathcal{W}|} A_{|\mathcal{W}|} - B_1 A_0 \\ &= \sum_{w=1}^{|\mathcal{W}|-1} (B_w - B_{w+1}) A_w + B_{|\mathcal{W}|} A_{|\mathcal{W}|} \end{aligned}$$

Thus we see (note there is a typo in the original proof)

$$\begin{aligned}
& \left| \sum_{w \in \mathcal{W}} J(a_{\bar{\pi}, w}, \eta(w))(A_{\bar{\pi}, w} - A_{\bar{\pi}, w-1}) \right| \\
&= \left| \left(\sum_{w=1}^{|\mathcal{W}|-1} [J(a_{\bar{\pi}, w}, \eta(w)) - J(a_{\bar{\pi}, w+1}, \eta(w+1))] A_{\bar{\pi}, w} \right) + J(a_{\bar{\pi}, |\mathcal{W}|}, \eta(|\mathcal{W}|)) A_{\bar{\pi}, |\mathcal{W}|} \right| \\
&\leq \left(\sum_{w=1}^{|\mathcal{W}|-1} |J(a_{\bar{\pi}, w}, \eta(w)) - J(a_{\bar{\pi}, w+1}, \eta(w+1))| \cdot |A_{\bar{\pi}, w}| \right) + |J(a_{\bar{\pi}, |\mathcal{W}|}, \eta(|\mathcal{W}|))| \cdot |A_{\bar{\pi}, |\mathcal{W}|}| \\
&\leq \left[\left(\sum_{w=1}^{|\mathcal{W}|-1} |J(a_{\bar{\pi}, w}, \eta(w)) - J(a_{\bar{\pi}, w+1}, \eta(w+1))| \right) + |J(a_{\bar{\pi}, |\mathcal{W}|}, \eta(|\mathcal{W}|))| \right] \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi}, v}| \\
&\leq \left[\left(\sum_{w=1}^{|\mathcal{W}|-1} |a_{\bar{\pi}, w+1} - a_{\bar{\pi}, w}| + |\eta(w+1) - \eta(w)| \right) + |J(a_{\bar{\pi}, |\mathcal{W}|}, \eta(|\mathcal{W}|))| \right] \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi}, v}| \\
&= \left[\left(\sum_{w=1}^{|\mathcal{W}|-1} P_{\bar{\pi}(w+1)} + \frac{1}{|\mathcal{W}| - 1} \right) + |J(a_{\bar{\pi}, |\mathcal{W}|}, \eta(|\mathcal{W}|))| \right] \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi}, v}| \\
&= [(1 - P_{\bar{\pi}(1)} + 1) + |J(1, 1)|] \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi}, v}| \\
&\leq (2 + |J(1, 1)|) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi}, v}| \\
&= O(1) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi}, v}|
\end{aligned}$$

as

$$|A_{\bar{\pi}, w}| \leq \sup_{v \in \mathcal{W}} |A_{\bar{\pi}, v}|$$

Back to (41), first note that

$$\mathbb{E}_{\bar{\pi}} I_{\bar{\pi}, w} = \mathbb{P}_{\bar{\pi}}(\bar{\pi}^{-1}(1) > w) = \frac{|\mathcal{W}| - w}{|\mathcal{W}|}$$

Thus

$$\begin{aligned}
& \left| \mathbb{E}_{\bar{\pi}} \left[\sum_{w \in \mathcal{W}} J(a_{\bar{\pi}, w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi}, w} \right] - \sum_{w \in \mathcal{W}} J\left(\frac{w\Delta}{|\mathcal{W}| - 1}, \eta(w)\right) \frac{|\mathcal{W}| - w}{|\mathcal{W}| - 1} \frac{\Delta}{|\mathcal{W}|} \right| \\
&= \left| \mathbb{E}_{\bar{\pi}} \left[\sum_{w \in \mathcal{W}} \left[J(a_{\bar{\pi}, w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi}, w} - J\left(\frac{w\Delta}{|\mathcal{W}| - 1}, \eta(w)\right) \frac{\Delta I_{\bar{\pi}, w}}{|\mathcal{W}| - 1} \right] \right] \right|
\end{aligned}$$

Using what we have deduced,

$$\begin{aligned}
& \left| \mathbb{E}_{\bar{\pi}} \left[\sum_{w \in \mathcal{W}} \left[J(a_{\bar{\pi},w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi},w} - J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w)\right) \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \right] \right] \right| \\
& \leq \mathbb{E}_{\bar{\pi}} \left| \left(\sum_{w \in \mathcal{W}} \left[J(a_{\bar{\pi},w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi},w} - J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w)\right) \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \right] \right) \right| \\
& \leq \mathbb{E}_{\bar{\pi}} \left[\left| \sum_{w \in \mathcal{W}} J(a_{\bar{\pi},w}, \eta(w)) \left(P_{\bar{\pi}(w)} - \frac{\Delta}{|\mathcal{W}|-1} \right) I_{\bar{\pi},w} \right| + O(1) \sum_{w \in \mathcal{W}} \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& = \mathbb{E}_{\bar{\pi}} \left[\left| \sum_{w \in \mathcal{W}} J(a_{\bar{\pi},w}, \eta(w)) \left(P_{\bar{\pi}(w)} - \frac{\Delta}{|\mathcal{W}|-1} \right) I_{\bar{\pi},w} \right| \right] + \mathbb{E}_{\bar{\pi}} \left[O(1) \sum_{w \in \mathcal{W}} \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& \leq \mathbb{E}_{\bar{\pi}} [O(1) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] + \mathbb{E}_{\bar{\pi}} \left[O(1) \sum_{w \in \mathcal{W}} \frac{\Delta I_{\bar{\pi},w}}{|\mathcal{W}|-1} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& \leq \mathbb{E}_{\bar{\pi}} [O(1) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] + \mathbb{E}_{\bar{\pi}} \left[O(1) \frac{\Delta}{|\mathcal{W}|-1} \left(\sum_{w \in \mathcal{W}} I_{\bar{\pi},w} \right) \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& = \mathbb{E}_{\bar{\pi}} [O(1) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] + \mathbb{E}_{\bar{\pi}} \left[O(1) \frac{\Delta}{|\mathcal{W}|-1} (\bar{\pi}^{-1}(1) - 1) \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& \leq \mathbb{E}_{\bar{\pi}} [O(1) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] + \mathbb{E}_{\bar{\pi}} \left[O(1) \frac{\Delta}{|\mathcal{W}|-1} (|\mathcal{W}|-1) \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& = \mathbb{E}_{\bar{\pi}} [O(1) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] + \mathbb{E}_{\bar{\pi}} \left[O(1) \frac{\Delta}{|\mathcal{W}|-1} (|\mathcal{W}|-1) \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& \leq \mathbb{E}_{\bar{\pi}} [O(1) \cdot \sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] + \mathbb{E}_{\bar{\pi}} \left[O(1) \sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \\
& = O(1) \cdot \left(\mathbb{E}_{\bar{\pi}} [\sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] + \mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \right) \\
& = O(1) \cdot 2 \cdot \left(\mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \right) \\
& = O(1) \cdot \left(\mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \right)
\end{aligned}$$

as we showed

$$\mathbb{E}_{\bar{\pi}} [\sup_{v \in \mathcal{W}} |A_{\bar{\pi},v}|] = \mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right]$$

in the proof of Lemma C.4.

Now by Lemma C.4,

$$\mathbb{E}_{\bar{\pi}} \left[\sup_{w \in \mathcal{W}} I_{\bar{\pi},w} \left| a_{\bar{\pi},w} - \frac{w\Delta}{|\mathcal{W}|-1} \right| \right] \leq \frac{1}{|\mathcal{W}|} + \sqrt{\left(\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \right) c_0 \log(c_0 |\mathcal{W}|) + P_{(2)} c_0 \log(c_0 |\mathcal{W}|)}$$

Since

$$P_{(i)} \leq P_{(2)}, \quad i \geq 2$$

then

$$\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 \leq P_{(2)} \sum_{i=2}^{|\mathcal{W}|} P_{(i)} = P_{(2)} \Delta \leq P_{(2)}$$

Thus

$$\sqrt{\sum_{i=2}^{|\mathcal{W}|} P_{(i)}^2 c_0 \log(c_0 |\mathcal{W}|)} = O\left(\sqrt{P_{(2)} \log |\mathcal{W}|}\right)$$

Since by assumption

$$P_{(2)} \log |\mathcal{W}| \longrightarrow 0$$

then for $|\mathcal{W}|$ large enough,

$$P_{(2)} \log |\mathcal{W}| \leq 1$$

and thus

$$P_{(2)} \log |\mathcal{W}| \leq \sqrt{P_{(2)} \log |\mathcal{W}|}$$

Therefore

$$\begin{aligned} \mathbb{E}_{\pi} \left[\sup_w I_{\pi,w} \left| a_{\pi,w} - \frac{w\Delta}{N-1} \right| \right] &= O\left(\frac{1}{|\mathcal{W}|} + \sqrt{P_{(2)} \log |\mathcal{W}|} \right) \\ &= O(1) \cdot \left(\frac{1}{|\mathcal{W}|} + \sqrt{P_{(2)} \log |\mathcal{W}|} \right) \end{aligned}$$

And so we have shown (42)

$$\begin{aligned} &\left| \mathbb{E}_{\bar{\pi}} \left[\sum_{w \in \mathcal{W}} J(a_{\bar{\pi},w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi},w} \right] - \sum_{w \in \mathcal{W}} J\left(\frac{w\Delta}{|\mathcal{W}|-1}, \eta(w) \right) \frac{|\mathcal{W}| - w}{|\mathcal{W}| - 1} \frac{\Delta}{|\mathcal{W}|} \right| \\ &\leq O(1) \cdot \left(\frac{1}{|\mathcal{W}|} + \sqrt{P_{(2)} \log |\mathcal{W}|} \right) \end{aligned} \tag{42}$$

So under the assumption, we turn to evaluate

$$\sum_{w \in \mathcal{W}} J\left(\frac{w\Delta}{|\mathcal{W}| - 1}, \eta(w) \right) \frac{|\mathcal{W}| - w}{|\mathcal{W}| - 1} \frac{\Delta}{|\mathcal{W}|}$$

and we have gotten ride of $\bar{\pi}$.

Now we move onto (43).

Recall Riemman sum (on left end point),

$$\sum_{k=1}^n f\left(\frac{k-1}{n} \right) \cdot \frac{1}{n} \xrightarrow{n \rightarrow \infty} \int_0^1 f(x) dx$$

This suggests how should we handle our problem.

Put aside Δ and consider

$$\sum_{w \in \mathcal{W}} J\left(\frac{w\Delta}{|\mathcal{W}| - 1}, \eta(w) \right) \frac{|\mathcal{W}| - w}{|\mathcal{W}| - 1} \cdot \frac{1}{|\mathcal{W}|}$$

Recall

$$\eta(w) = \frac{w-1}{|\mathcal{W}| - 1}$$

then we observe

$$\frac{N-w}{N-1} = 1 - \eta(w)$$

and

$$\frac{w\Delta}{N-1} = \eta(w)\Delta + \frac{\Delta}{N-1}$$

Thus we obtain

$$\begin{aligned} &\sum_{w \in \mathcal{W}} J\left(\frac{w\Delta}{|\mathcal{W}| - 1}, \eta(w) \right) \frac{|\mathcal{W}| - w}{|\mathcal{W}| - 1} \frac{1}{|\mathcal{W}|} \\ &= \sum_{w \in \mathcal{W}} J\left(\eta(w)\Delta + \frac{\Delta}{|\mathcal{W}| - 1}, \eta(w) \right) (1 - \eta(w)) \frac{1}{|\mathcal{W}|} \end{aligned}$$

This natrually suggests we should consider

$$\sum_{w \in \mathcal{W}} J(\eta(w)\Delta, \eta(w))(1 - \eta(w)) \frac{1}{|\mathcal{W}|}$$

Then

$$\begin{aligned} & \left| \left[\sum_{w \in \mathcal{W}} J\left(\eta(w)\Delta + \frac{\Delta}{|\mathcal{W}| - 1}, \eta(w)\right)(1 - \eta(w)) \frac{1}{|\mathcal{W}|} \right] - \left[\sum_{w \in \mathcal{W}} J(\eta(w)\Delta, \eta(w))(1 - \eta(w)) \frac{1}{|\mathcal{W}|} \right] \right| \\ &= \frac{1}{|\mathcal{W}|} \left| \sum_{w \in \mathcal{W}} \left(J\left(\eta(w)\Delta + \frac{\Delta}{|\mathcal{W}| - 1}, \eta(w)\right) - J(\eta(w)\Delta, \eta(w)) \right) (1 - \eta(w)) \right| \\ &\leq \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \left| \left(J\left(\eta(w)\Delta + \frac{\Delta}{|\mathcal{W}| - 1}, \eta(w)\right) - J(\eta(w)\Delta, \eta(w)) \right) (1 - \eta(w)) \right| \\ &\leq \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \left| \eta(w)\Delta + \frac{\Delta}{|\mathcal{W}| - 1} - \eta(w)\Delta \right| (1 - \eta(w)) \\ &= \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \left(\frac{\Delta}{|\mathcal{W}| - 1} \right) (1 - \eta(w)) \\ &\leq \left[\frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \left(\frac{\Delta}{|\mathcal{W}| - 1} \right) \right] \\ &= \frac{\Delta}{|\mathcal{W}| - 1} \\ &= \frac{O(1)}{|\mathcal{W}|} \end{aligned}$$

Looking at

$$\sum_{w \in \mathcal{W}} J(\eta(w)\Delta, \eta(w))(1 - \eta(w)) \frac{1}{|\mathcal{W}|}$$

Since

$$\eta(w) = \frac{w - 1}{|\mathcal{W}| - 1}$$

then for $w \in \mathcal{W}$, $\eta(w)$ takes value on $\{0, \frac{1}{|\mathcal{W}| - 1}, \dots, \frac{|\mathcal{W}| - 2}{|\mathcal{W}| - 1}, 1\}$.

So this is a Riemann sum for

$$\int_0^1 J(x\Delta, x)(1 - x) dx$$

Precisely, let

$$I_w := \left[\frac{w - 1}{|\mathcal{W}|}, \frac{w}{|\mathcal{W}|} \right]$$

let

$$f(x) := J(x\Delta, x)(1 - x)$$

and let

$$M := \sup_{(a,b) \in [0,1]^2} |J(a,b)| < \infty$$

If $x, y \in [0, 1]$, add and subtract the cross term $(1 - y)J(\Delta x, x)$ gives

$$\begin{aligned} f(y) - f(x) &= (1 - y)J(\Delta y, y) - (1 - x)J(\Delta x, x) \\ &= (1 - y)[J(\Delta y, y) - J(\Delta x, x)] + (x - y)J(\Delta x, x) \end{aligned}$$

Thus

$$|f(y) - f(x)| \leq (1 - y)|J(\Delta y, y) - J(\Delta x, x)| + |x - y| |J(\Delta x, x)|$$

By Lipschitz,

$$\begin{aligned}
|J(\Delta y, y) - J(\Delta x, x)| &\leq |\Delta y - \Delta x| + |y - x| \\
&= (\Delta + 1)|y - x| \\
&\leq 2|y - x|
\end{aligned}$$

By boundness,

$$|J(\Delta x, x)| \leq M$$

Thus

$$\begin{aligned}
|f(y) - f(x)| &\leq 2(1 - y)|y - x| + M|y - x| \\
&\leq (M + 2)|y - x|
\end{aligned}$$

Since $\eta(w) \in I_w$, then for any $x \in I_w$,

$$|\eta(w) - x| \leq \frac{1}{|\mathcal{W}|}$$

so for any $x \in I_w$,

$$|f(\eta(w)) - f(x)| \leq \frac{M + 2}{|\mathcal{W}|}$$

Then

$$\begin{aligned}
&\left| \sum_{w=1}^{|\mathcal{W}|} J(\eta(w)\Delta, \eta(w))(1 - \eta(w)) \frac{1}{|\mathcal{W}|} - \int_0^1 f(x) dx \right| \\
&= \left| \sum_{w=1}^{|\mathcal{W}|} \int_{I_w} (f(\eta(w)) - f(x)) dx \right| \\
&\leq \sum_{w=1}^{|\mathcal{W}|} \int_{I_w} |f(\eta(w)) - f(x)| dx \\
&\leq \sum_{w=1}^{|\mathcal{W}|} \frac{M + 2}{|\mathcal{W}|} \cdot \frac{1}{|\mathcal{W}|} \\
&= \frac{M + 2}{|\mathcal{W}|} \\
&= \frac{O(1)}{|\mathcal{W}|}
\end{aligned}$$

Therefore by triangle inequality,

$$\begin{aligned}
&\left| \left[\sum_{w \in \mathcal{W}} J\left(\frac{w\Delta}{|\mathcal{W}| - 1}, \eta(w)\right) \frac{|\mathcal{W}| - w}{|\mathcal{W}| - 1} \frac{1}{|\mathcal{W}|} \right] - \int_0^1 f(x) dx \right| \\
&= \left| \left[\sum_{w \in \mathcal{W}} J\left(\eta(w)\Delta + \frac{\Delta}{|\mathcal{W}| - 1}, \eta(w)\right) (1 - \eta(w)) \frac{1}{|\mathcal{W}|} \right] - \int_0^1 f(x) dx \right| \\
&\leq \frac{O(1)}{|\mathcal{W}|} + \frac{O(1)}{|\mathcal{W}|} \\
&= \frac{O(1)}{|\mathcal{W}|}
\end{aligned}$$

And so we have shown (43).

Finally we can take a look at what we got.

Denote

$$M_{|\mathcal{W}|} := \mathbb{E}_{\bar{\pi}} \sum_{w \in \mathcal{W}} J(a_{\bar{\pi}, w}, \eta(w)) P_{\bar{\pi}(w)} I_{\bar{\pi}, w}$$

and

$$S_{|\mathcal{W}|} := \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} J\left(\frac{w\Delta}{|\mathcal{W}| - 1}, \eta(w)\right) \frac{|\mathcal{W}| - w}{|\mathcal{W}| - 1}$$

and

$$I_{\Delta} := \int_0^1 (1-x)J(\Delta x, x) dx$$

In (40), we showed

$$|\mathcal{T}_1 - M_{|\mathcal{W}|}| \leq C_1 \cdot P_{(2)}$$

In (42), we showed

$$|M_{|\mathcal{W}|} - \Delta S_{|\mathcal{W}|}| \leq C_2 \cdot \left(\frac{1}{|\mathcal{W}|} + \sqrt{P_{(2)} \log |\mathcal{W}|} \right)$$

In (43), we showed

$$|S\Delta - I_{\Delta}\Delta| \leq \frac{C_3 \cdot \Delta}{|\mathcal{W}|} \leq C_3 \cdot \frac{1}{|\mathcal{W}|}$$

By triangle inequality,

$$\begin{aligned} |\mathcal{T}_1 - \Delta I_{\Delta}| &\leq C_1 P_{(2)} + C_2 \left(\frac{1}{|\mathcal{W}|} + \sqrt{P_{(2)} \log |\mathcal{W}|} \right) + C_3 \frac{1}{|\mathcal{W}|} \\ &\leq C \cdot \left(P_{(2)} + \frac{1}{|\mathcal{W}|} + \sqrt{P_{(2)} \log |\mathcal{W}|} \right) \end{aligned}$$

Recall the definition for $\mathcal{Q}_{\Delta}$, for every $\mathbf{P} \in \mathcal{Q}_{\Delta}$,

$$P_{(2)} \log |\mathcal{W}| \leq \varepsilon_{|\mathcal{W}|}, \quad \varepsilon_{|\mathcal{W}|} \xrightarrow{|\mathcal{W}| \rightarrow \infty} 0$$

Therefore

$$|\mathcal{T}_1 - \Delta I| \xrightarrow{|\mathcal{W}| \rightarrow \infty} 0$$

uniformly over $\mathbf{P} \in \mathcal{Q}_{\Delta}$ and $\Delta \in \left[0, 1 - \frac{1}{|\mathcal{W}|}\right]$.

That is,

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{J \in \mathcal{J}} \sup_{\Delta \in \left[0, 1 - \frac{1}{|\mathcal{W}|}\right]} \sup_{\mathbf{P} \in \mathcal{Q}_{\Delta}} \left| \mathcal{T}_1 - \Delta \int_0^1 (1-x)J(\Delta x, x) dx \right| = 0$$

Consider $\mathcal{T}_3$ (the case where $l = w$).

Consider

$$\begin{aligned} \mathcal{T}_3 &= \frac{1}{|\mathcal{W}|!} \sum_{w=1}^{|\mathcal{W}|} \sum_{\pi \in \Pi_w} \int_{a_{\pi, w-1}}^{a_{\pi, w}} J(r, \eta(w)) dr \\ &= \mathbb{E}_{\bar{\pi}} \sum_{w=1}^{|\mathcal{W}|} \int_{a_{\bar{\pi}, w-1}}^{a_{\bar{\pi}, w}} J(r, \eta(w)) \mathbf{1}_{\bar{\pi}(w)=1} dr \\ &= \frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} \mathbb{E}_{\bar{\pi}} \left[\int_{a_{\bar{\pi}, w-1}}^{a_{\bar{\pi}, w-1}+1-\Delta} J(r, \eta(w)) dr \mid \bar{\pi}(w) = 1 \right] \end{aligned}$$

The first equality is clear using previous logics.

For the second equality, for a fixed w , denote

$$X_w := \int_{a_{\bar{\pi}, w-1}}^{a_{\bar{\pi}, w}} J(r, \eta(w)) dr$$

By definition,

$$\mathbb{E}[X_w \mathbf{1}_{\{\bar{\pi}(w)=1\}}] = \mathbb{P}(\bar{\pi}(w) = 1) \mathbb{E}[X_w \mid \bar{\pi}(w) = 1]$$

Observe

$$\mathbb{P}(\bar{\pi}(w) = 1) = \frac{(|\mathcal{W}| - 1)!}{|\mathcal{W}|!} = \frac{1}{|\mathcal{W}|}$$

Note on the event $\{\bar{\pi}(w) = 1\}$,

$$P_{\bar{\pi}(w)} = P_1 = 1 - \Delta$$

By definition,

$$a_{\bar{\pi},w} = a_{\bar{\pi},w-1} + P_{\bar{\pi}(w)}$$

Thus condition on $\bar{\pi}(w) = 1$,

$$a_{\bar{\pi},w} = a_{\bar{\pi},w-1} + 1 - \Delta.$$

Therefore

$$\mathbb{E}_{\bar{\pi}} [X_w \mathbf{1}_{\bar{\pi}(w)=1}] = \frac{1}{|\mathcal{W}|} \cdot \mathbb{E}_{\bar{\pi}} \left[\int_{a_{\bar{\pi},w-1}}^{a_{\bar{\pi},w-1}+1-\Delta} J(r, \eta(w)) dr \middle| \bar{\pi}(w) = 1 \right]$$

which is what we want.

For (44), the idea is to use $[\eta(w)\Delta, \eta(w)\Delta + 1 - \Delta]$ to approximate $[a_{\bar{\pi},w-1}, a_{\bar{\pi},w-1} + 1 - \Delta]$, because

$$\begin{aligned} \mathbb{E}_{\bar{\pi}} [a_{\bar{\pi},w-1} \mid \bar{\pi}(w) = 1] &= \frac{w-1}{|\mathcal{W}|-1} \sum_{i=2}^{|\mathcal{W}|} P_{(i)} \\ &= \frac{w-1}{|\mathcal{W}|-1} \Delta \\ &= \eta(w)\Delta \end{aligned}$$

Denote

$$c := 1 - \Delta$$

and

$$M := \sup_{(a,b) \in [0,1]^2} |J(a,b)| < \infty$$

If $a \leq b$, then

$$\int_a^{a+c} J(r, x) dr - \int_b^{b+c} J(r, x) dr = \int_a^b J(r, x) dr - \int_{a+c}^{b+c} J(r, x) dr$$

Thus

$$\begin{aligned} \left| \int_a^{a+c} J(r, x) dr - \int_b^{b+c} J(r, x) dr \right| &\leq \int_a^b |J(r, x)| dr + \int_{a+c}^{b+c} |J(r, x)| dr \\ &\leq 2M|a-b| \end{aligned}$$

The case is similar for $a > b$.

Therefore

$$\left| \int_a^{a+c} J(r, x) dr - \int_b^{b+c} J(r, x) dr \right| \leq 2M|a-b|$$

By triangle inequality, note $\eta(w) = \frac{w-1}{|\mathcal{W}|-1}$,

$$\begin{aligned}
& \left| \mathcal{T}_3 - \frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} \int_{\eta(w)\Delta}^{\eta(w)\Delta+1-\Delta} J(r, \eta(w)) dr \right| \\
&= \left| \frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} \mathbb{E}_{\bar{\pi}} \left[\int_{a_{\bar{\pi}, w-1}}^{a_{\bar{\pi}, w-1}+1-\Delta} J(r, \eta(w)) dr \right] \pi(w) = 1 \right] - \frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} \int_{\eta(w)\Delta}^{\eta(w)\Delta+1-\Delta} J(r, \eta(w)) dr \right| \\
&\leq \frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} \mathbb{E}_{\bar{\pi}} \left[\left| \int_{a_{\bar{\pi}, w-1}}^{a_{\bar{\pi}, w-1}+1-\Delta} J(r, \eta(w)) dr - \int_{\eta(w)\Delta}^{\eta(w)\Delta+1-\Delta} J(r, \eta(w)) dr \right| \middle| \pi(w) = 1 \right] \\
&\leq \frac{2M}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} \mathbb{E}_{\bar{\pi}} [|a_{\bar{\pi}, w-1} - \eta(w)\Delta| | \pi(w) = 1]
\end{aligned}$$

Using

$$\frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} x_w \leq \sup_{w \in |\mathcal{W}|} x_w$$

we get

$$\begin{aligned}
& \left| \mathcal{T}_3 - \frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} \int_{\frac{w-1}{|\mathcal{W}|-1}\Delta}^{\frac{w-1}{|\mathcal{W}|-1}\Delta+1-\Delta} J(r, \eta(w)) dr \right| \\
&\leq O(1) \cdot \sup_{w \in \mathcal{W}} \mathbb{E}_{\bar{\pi}} \left[\left| a_{\bar{\pi}, w-1} - \frac{w-1}{|\mathcal{W}|-1}\Delta \right| \middle| \pi(w) = 1 \right] \\
&\leq O(1) \cdot \left(\frac{1}{|\mathcal{W}|} + \sqrt{P_{(2)} \log |\mathcal{W}|} \right)
\end{aligned}$$

using Lemma C.4 and similar logic in $\mathcal{T}_1$ case.

Thus we have shown (44).

For (45), similarly we turn to Riemann sum.

Define

$$G_{\Delta}(x) := \int_{x\Delta}^{x\Delta+1-\Delta} J(r, x) dr$$

Let $s := r - x\Delta$, then

$$G_{\Delta}(x) = \int_0^{1-\Delta} J(x\Delta + s, x) ds$$

Let $x, y \in [0, 1]$, then by Lipschitz

$$\begin{aligned}
|G_{\Delta}(x) - G_{\Delta}(y)| &= \left| \int_0^{1-\Delta} [J(\Delta x + s, x) - J(\Delta y + s, y)] ds \right| \\
&\leq \int_0^{1-\Delta} |J(\Delta x + s, x) - J(\Delta y + s, y)| ds \\
&\leq \int_0^{1-\Delta} |\Delta x + s - (\Delta y + s)| + |x - y| ds \\
&= \int_0^{1-\Delta} (\Delta + 1) |x - y| ds \\
&= (1 - \Delta)(1 + \Delta) |x - y| \\
&= (1 - \Delta^2) |x - y| \\
&\leq |x - y|
\end{aligned}$$

Define

$$I_w := \left[\frac{w-1}{|\mathcal{W}|}, \frac{w}{|\mathcal{W}|} \right]$$

Since $\eta(w) \in I_w$, then for any $x \in I_w$,

$$|\eta(w) - x| \leq \frac{1}{|\mathcal{W}|}$$

Thus for any $x \in I_w$,

$$|G_\Delta(\eta(w)) - G_\Delta(x)| \leq \frac{1}{|\mathcal{W}|}$$

Therefore

$$\begin{aligned} & \left| \frac{1}{|\mathcal{W}|} \sum_{w=1}^{|\mathcal{W}|} G_\Delta(\eta(w)) - \int_0^1 G_\Delta(x) dx \right| \\ &= \left| \sum_{w=1}^{|\mathcal{W}|} \int_{I_w} [G_\Delta(\eta(w)) - G_\Delta(x)] dx \right| \\ &\leq \sum_{w=1}^{|\mathcal{W}|} \int_{I_w} \frac{1}{|\mathcal{W}|} dx \\ &= |\mathcal{W}| \cdot \frac{1}{|\mathcal{W}|} \cdot \frac{1}{|\mathcal{W}|} \\ &= \frac{1}{|\mathcal{W}|} \end{aligned}$$

Therefore we have shown (45).

The rest is similar.

□

Corollary C.1 and Theorem 4.1

Note the result for "bounded and measurable J " is wrong, there are counterexamples.

But note in Theorem 4.1, we only used

$$J_r(u, x) = \mathbf{1}_{\{|u-x| \leq r\}}$$

We will prove this version holds.

Proof.

Define

$$N := |\mathcal{W}|$$

and

$$Z_N := (U_t, \eta_N(\pi_t(w_t)))$$

Let $\mathbf{P}_{|\mathcal{W}|}$ be a sequence of NTP distributions satisfying

$$P_{|\mathcal{W}|,(1)} \longrightarrow 1 - \Delta, \quad P_{|\mathcal{W}|,(2)} \log |\mathcal{W}| \longrightarrow 0$$

and define

$$\Delta_N := 1 - P_{N,(1)}.$$

Thus,

$$\Delta_N \rightarrow \Delta$$

Assume $0 \leq \Delta < 1$ and fix

$$r \in [0, 1 - \Delta]$$

For any bounded function $F : [0, 1]^2 \rightarrow \mathbb{R}$, define

$$\begin{aligned}\mathcal{L}_\delta(F) &:= \delta \int_0^1 (1-x) F(\delta x, x) dx \\ &\quad + \delta \int_0^1 x F(\delta x + 1 - \delta, x) dx \\ &\quad + \int_0^1 dy \int_{\delta y}^{\delta y + 1 - \delta} F(u, y) du\end{aligned}$$

We shall prove that

$$\mathbb{E}_{1, P_N} [J_r(Z_N)] \longrightarrow \mathcal{L}_\Delta(J_r)$$

Let

$$d(u, x) := |u - x|$$

and define the clipping operator

$$[z]_{[0,1]} := \min\{1, \max\{z, 0\}\}$$

For every $\varepsilon > 0$, define

$$J_{r,\varepsilon}^-(u, x) := \left[\frac{r - d(u, x)}{\varepsilon} \right]_{[0,1]}$$

and

$$J_{r,\varepsilon}^+(u, x) := \left[1 + \frac{r - d(u, x)}{\varepsilon} \right]_{[0,1]}$$

Equivalently,

$$J_{r,\varepsilon}^-(u, x) = \begin{cases} 1, & d(u, x) \leq r - \varepsilon \\ \frac{r - d(u, x)}{\varepsilon}, & r - \varepsilon < d(u, x) < r \\ 0, & d(u, x) \geq r \end{cases}$$

whereas

$$J_{r,\varepsilon}^+(u, x) = \begin{cases} 1, & d(u, x) \leq r \\ 1 - \frac{r - d(u, x)}{\varepsilon}, & r < d(u, x) < r + \varepsilon \\ 0, & d(u, x) \geq r + \varepsilon \end{cases}$$

These functions satisfy

$$J_{r,\varepsilon}^- \leq J_r \leq J_{r,\varepsilon}^+$$

Moreover,

$$||u - x| - |v - y|| \leq |u - v| + |x - y|$$

so $J_{r,\varepsilon}^-$ and $J_{r,\varepsilon}^+$ are $O(\varepsilon^{-1})$ -Lipschitz.

Although Lemma C.3 is stated for 1-Lipschitz functions, it also applies to any fixed Lipschitz function by rescaling. Hence, for every fixed $\varepsilon > 0$,

$$\mathbb{E}_{1, P_N} [J_{r,\varepsilon}^\pm(Z_N)] - \mathcal{L}_{\Delta_N}(J_{r,\varepsilon}^\pm) \longrightarrow 0.$$

For every fixed bounded continuous F ,

$$\mathcal{L}_{\Delta_N}(F) \longrightarrow \mathcal{L}_\Delta(F).$$

Indeed, the first two terms follow directly from the dominated convergence theorem. For the third term, write

$$\int_{\Delta_N y}^{\Delta_N y + 1 - \Delta_N} F(u, y) du = \int_0^{1 - \Delta_N} F(\Delta_N y + s, y) ds$$

and apply the dominated convergence theorem.

Consequently, for every fixed $\varepsilon > 0$,

$$\mathbb{E}_{1, \mathbf{P}_N} [J_{r, \varepsilon}^{\pm}(Z_N)] \longrightarrow \mathcal{L}_{\Delta} (J_{r, \varepsilon}^{\pm})$$

Since

$$J_{r, \varepsilon}^{-} \leq J_r \leq J_{r, \varepsilon}^{+}$$

we have

$$\mathbb{E} [J_{r, \varepsilon}^{-}(Z_N)] \leq \mathbb{E}[J_r(Z_N)] \leq \mathbb{E} [J_{r, \varepsilon}^{+}(Z_N)]$$

Taking $N \rightarrow \infty$ while keeping ε fixed gives

$$\begin{aligned} \mathcal{L}_{\Delta} (J_{r, \varepsilon}^{-}) &\leq \liminf_{N \rightarrow \infty} \mathbb{E}[J_r(Z_N)] \\ &\leq \limsup_{N \rightarrow \infty} \mathbb{E}[J_r(Z_N)] \\ &\leq \mathcal{L}_{\Delta} (J_{r, \varepsilon}^{+}) \end{aligned}$$

It remains to prove that the gap between the upper and lower bounds tends to zero as $\varepsilon \downarrow 0$.

The functions $J_{r, \varepsilon}^{+}$ and $J_{r, \varepsilon}^{-}$ can differ only on the boundary layer

$$B_{r, \varepsilon} := \{(u, x) : ||u - x| - r| \leq \varepsilon\}$$

Therefore,

$$0 \leq J_{r, \varepsilon}^{+} - J_{r, \varepsilon}^{-} \leq \mathbf{1}_{B_{r, \varepsilon}}$$

and hence

$$\begin{aligned} 0 &\leq \mathcal{L}_{\Delta} (J_{r, \varepsilon}^{+}) - \mathcal{L}_{\Delta} (J_{r, \varepsilon}^{-}) \\ &\leq \mathcal{L}_{\Delta} (\mathbf{1}_{B_{r, \varepsilon}}) \end{aligned}$$

We now bound the three terms in the last expression. Set

$$c := 1 - \Delta > 0$$

For the first term in $\mathcal{L}_{\Delta}$, the relevant point is

$$(u, x) = (\Delta x, x)$$

and therefore

$$|u - x| = cx$$

It follows that

$$\begin{aligned} &\Delta \int_0^1 (1 - x) \mathbf{1}_{|cx - r| \leq \varepsilon} dx \\ &\leq \lambda \{x \in [0, 1] : |cx - r| \leq \varepsilon\} \\ &\leq \frac{2\varepsilon}{c} \end{aligned}$$

For the second term,

$$(u, x) = (\Delta x + c, x)$$

so

$$|u - x| = c(1 - x)$$

Thus,

$$\Delta \int_0^1 x \mathbf{1}_{|c(1-x) - r| \leq \varepsilon} dx \leq \frac{2\varepsilon}{c}$$

For the third term, fix x . The condition

$$||u - x| - r| \leq \varepsilon$$

restricts u to at most two intervals, each of length at most 2ε . Hence their total length is at most 4ε , and therefore

$$\int_0^1 dx \int_{\Delta x}^{\Delta x+c} \mathbf{1}_{||u-x|-r|\leq\varepsilon} du \leq 4\varepsilon$$

Combining these estimates yields

$$\mathcal{L}_\Delta(\mathbf{1}_{B_{r,\varepsilon}}) \leq \frac{4\varepsilon}{c} + 4\varepsilon \longrightarrow 0$$

Consequently,

$$\mathcal{L}_\Delta(J_{r,\varepsilon}^+) - \mathcal{L}_\Delta(J_{r,\varepsilon}^-) \longrightarrow 0$$

Since

$$J_{r,\varepsilon}^- \leq J_r \leq J_{r,\varepsilon}^+$$

positivity of $\mathcal{L}_\Delta$ also gives

$$\mathcal{L}_\Delta(J_{r,\varepsilon}^-) \leq \mathcal{L}_\Delta(J_r) \leq \mathcal{L}_\Delta(J_{r,\varepsilon}^+)$$

Thus

$$\mathcal{L}_\Delta(J_{r,\varepsilon}^\pm) \longrightarrow \mathcal{L}_\Delta(J_r)$$

Therefore by monotone convergence and squeeze theorem,

$$\mathbb{E}_{1, P_N}[J_r(Z_N)] \longrightarrow \mathcal{L}_\Delta(J_r)$$

This proves the restricted extension of Lemma C.3 required in Theorem 4.1. □

It remains to compute $\mathcal{L}_\Delta(J_r)$. Recall that

$$c = 1 - \Delta, \quad a := \frac{r}{c} \in [0, 1]$$

Substituting J_r into $\mathcal{L}_\Delta$ gives

$$\begin{aligned} \mathcal{L}_\Delta(J_r) &= \Delta \int_0^1 (1-x) \mathbf{1}_{cx \leq r} dx \\ &\quad + \Delta \int_0^1 x \mathbf{1}_{c(1-x) \leq r} dx \\ &\quad + \int_0^1 dx \int_{\Delta x}^{\Delta x+c} \mathbf{1}_{|u-x| \leq r} du \end{aligned}$$

The first two terms are equal by the substitution $y = 1 - x$. Their sum is

$$\begin{aligned} 2\Delta \int_0^a (1-x) dx &= 2\Delta \left(a - \frac{a^2}{2} \right) \\ &= \Delta(2a - a^2) \end{aligned}$$

For the third term, fixing x and subtracting x from the endpoints transforms the interval

$$[\Delta x, \Delta x + c]$$

into

$$[-cx, c(1-x)].$$

Its intersection with $[-r, r]$ has length

$$c - (cx - r)_+ - (c(1-x) - r)_+$$

Therefore, the third term is

$$\begin{aligned}
& c - \int_0^1 (cx - r)_+ dx - \int_0^1 (c(1-x) - r)_+ dx \\
& = c - 2 \int_0^1 (cx - r)_+ dx
\end{aligned}$$

Since $r = ca$,

$$\begin{aligned}
\int_0^1 (cx - r)_+ dx &= \int_a^1 c(x - a) dx \\
&= \frac{c}{2}(1 - a)^2
\end{aligned}$$

Thus, the third term equals

$$c - c(1 - a)^2 = c(2a - a^2)$$

Combining all three terms and using $\Delta + c = 1$, we obtain

$$\begin{aligned}
\mathcal{L}_\Delta(J_r) &= \Delta(2a - a^2) + c(2a - a^2) \\
&= 2a - a^2 \\
&= 1 - (1 - a)^2
\end{aligned}$$

Substituting $a = r/(1 - \Delta)$ gives

$$\mathcal{L}_\Delta(J_r) = 1 - \left(1 - \frac{r}{1 - \Delta}\right)^2$$

Since

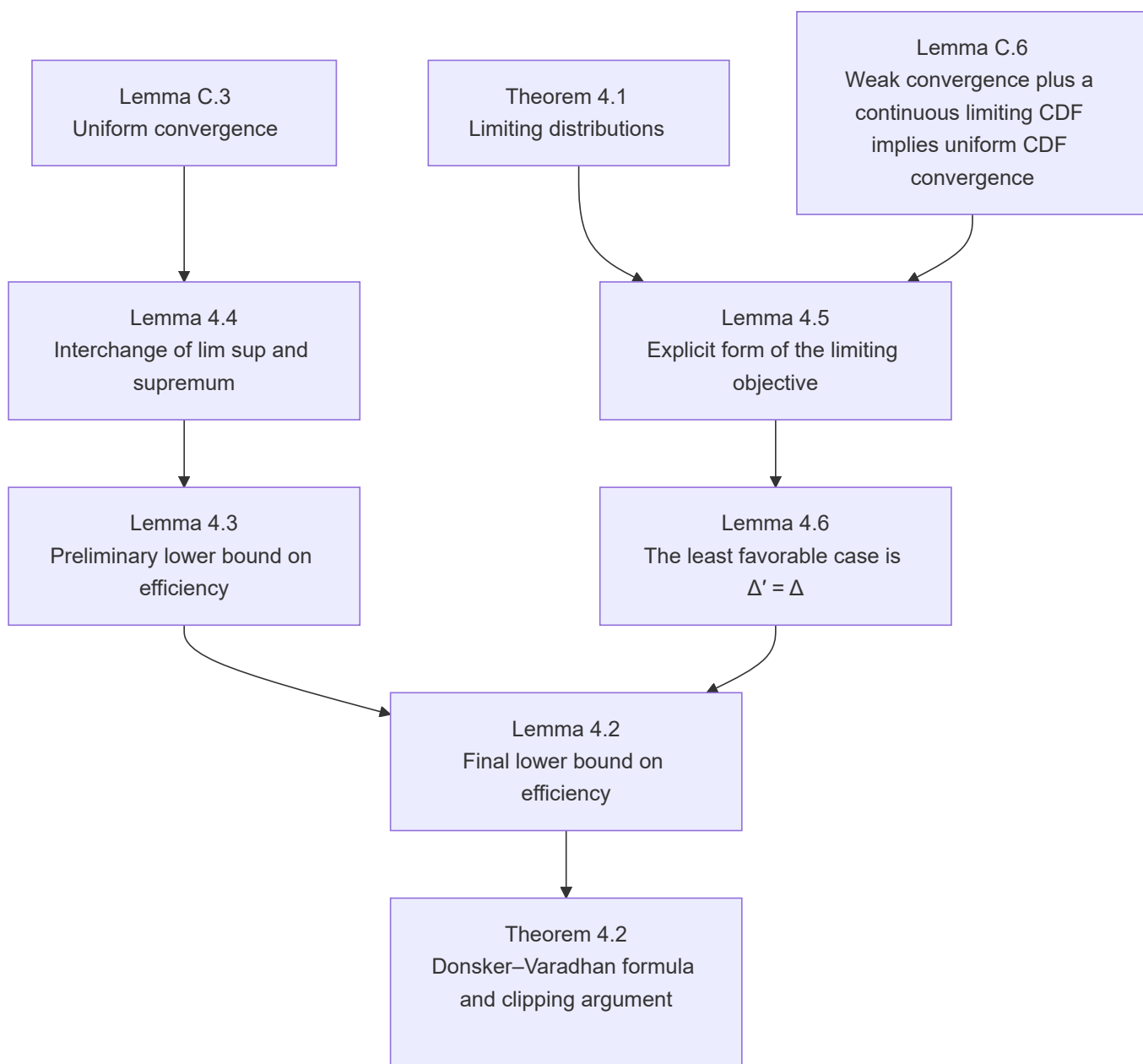
$$J_r(Z_N) = \mathbf{1}_{|U_t - \eta_N(\pi_t(w_t))| \leq r} = \mathbf{1}_{Y_t^{\text{dif}} \leq r}$$

we conclude that

$$\lim_{N \rightarrow \infty} \mathbb{P}_{H_1}(Y_t^{\text{dif}} \leq r \mid \mathbf{P}_N) = 1 - \left(1 - \frac{r}{1 - \Delta}\right)^2, \quad 0 \leq r \leq 1 - \Delta$$

which is Theorem 4.1.

Theorem 4.2



Where f is a continuous real-valued function of a real variable and v is a non-decreasing real function, the Lebesgue-Stieltjes integral is equivalent to the Riemann-Stieltjes integral, in which case we often write

$$\int_a^b f(x) dv(x)$$

This is the case in the paper.

Theorem 8 (Portmanteau Theorem).

Let X_n and X be real-valued random variables, and let $\mu_n = \mathcal{L}(X_n)$ and $\mu = \mathcal{L}(X)$ denote their probability laws. Then the following statements are equivalent:

- The probability measures μ_n converge weakly to μ ,

$$\mu_n \Rightarrow \mu$$

- X_n converges in distribution to X ,

$$X_n \xrightarrow{d} X$$

that is, at every continuity point x of the CDF F ,

$$F_n(x) \longrightarrow F(x)$$

where $F_n(x) = \mathbb{P}(X_n \leq x)$, $F(x) = \mathbb{P}(X \leq x)$.

- For every bounded continuous function $f : \mathbb{R} \mapsto \mathbb{R}$,

$$\mathbb{E}[f(X_n)] \longrightarrow \mathbb{E}[f(X)]$$

Lemma 9 (Lemma 4.4).

For any Lipschitz-continuous function h on $[0, 1]$, we have

$$\limsup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left[\mathbb{E}_0 h(Y^{\text{dif}}) + \log \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] \right] = \sup_{\Delta' \geq \Delta} \limsup_{P \in \mathcal{Q}_{\Delta'}} \left[\mathbb{E}_0 h(Y^{\text{dif}}) + \log \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] \right]$$

Proof.

Since under the null, $\mathbb{E}_0 h(Y^{\text{dif}})$ does not depend on $\mathbf{P}$ (and so on Δ'). By Theorem 4.1, the CDF converges pointwise for all $r \in [0, 1]$. By Portmanteau Theorem, since h is bounded and continuous on $[0, 1]$,

$$\lim_{|\mathcal{W}| \rightarrow \infty} \mathbb{E}_0 h(Y^{\text{dif}}) = \int_{[0,1]} h(r) F_{\text{dif},0}(dr)$$

Then it suffices to consider (with $\log$)

$$\phi_{P,h}(1) = \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right]$$

Take

$$J(x, y) = \exp(-h(|x - y|))$$

for $x, y \in [0, 1]$ and so

$$J(U_t, \eta(\pi_t(w_t))) = e^{-h(Y^{\text{dif}})}$$

Thus we can utilize Lemma C.3.

But first we check that J is indeed lipschitz. Consider $J(x, y)$ and $J(x', y')$.

Since h is L -lipschitz hence continuous, by extreme value theorem, $\min_{r \in [0,1]} h(r)$ exists. Also continuous function maps compact sets to compact sets (closed and bounded on $\mathbb{R}$). Thus

$$h([0, 1]) = \left[\min_{r \in [0,1]} h(r), \max_{r \in [0,1]} h(r) \right]$$

Apply mean value theorem between $h(|x - y|)$ and $h(|x' - y'|)$ for $g(z) := e^{-z}$. We get

$$e^{-h(|x-y|)} - e^{-h(|x'-y'|)} = -e^{-k} (h(|x - y|) - h(|x' - y'|))$$

for some $k \in [h(|x - y|), h(|x' - y'|)] \subseteq [\min_{r \in [0,1]} h(r), \max_{r \in [0,1]} h(r)]$, or
 $[h(|x' - y'|), h(|x - y|)] \subseteq [\min_{r \in [0,1]} h(r), \max_{r \in [0,1]} h(r)]$.

We note that

$$||x - y| - |x' - y'|| \leq |x - x'| + |y - y'|$$

This is because by reverse triangle inequality,

$$\begin{aligned} ||x - y| - |x' - y'|| &\leq |(x - y) - (x' - y')| \\ &= |(x - x') - (y - y')| \\ &\leq |x - x'| + |y - y'| \end{aligned}$$

Therefore

$$\begin{aligned}
|J(x, y) - J(x', y')| &= |e^{-h(|x-y|)} - e^{-h(|x'-y'|)}| \\
&\leq e^{-\min_{r \in [0,1]} h(r)} |h(|x-y|) - h(|x'-y'|)| \\
&\leq e^{-\min_{r \in [0,1]} h(r)} L ||x-y| - |x'-y'|| \\
&\leq e^{-\min_{r \in [0,1]} h(r)} L (|x-x'| + |y-y'|) \\
&\leq e^{-\min_{r \in [0,1]} h(r)} L ||(x, y) - (x', y')||_1 \\
&\leq e^{-\min_{r \in [0,1]} h(r)} \sqrt{2} L ||(x, y) - (x', y')||_2
\end{aligned}$$

The last inequality is because for any 2 p -norms in a finite-dimensional normed vector space, there exists constant c and C such that

$$c||z||_a \leq ||z||_b \leq C||z||_a$$

In particular, in our 2-dimensional case,

$$||z||_2 \leq ||z||_1 \leq \sqrt{2}||z||_2$$

Therefore we have shown that J is lipshcitz.

Next we show

$$\begin{aligned}
&\lim_{|\mathcal{W}| \rightarrow \infty} \left| \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) - \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right| \\
&\leq \lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right| \\
&= 0
\end{aligned}$$

First by Lemma C.3,

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right| = 0 \quad (46)$$

Next, the inequality is intuitively true, but here is the proof. Note

$$\begin{aligned}
\phi_{P,h}(1) &\leq \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) + \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right| \\
\sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) &\leq \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) + \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right|
\end{aligned}$$

thus

$$\sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) - \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \leq \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right|$$

Conversely, from

$$\lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \leq \phi_{P,h}(1) + \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right|$$

we can similarly get

$$\sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) - \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) \leq \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right|$$

Therefore,

$$\left| \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) - \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right| \leq \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right|$$

as desired.

Therefore, by definition of limit, it follows that

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) = \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \quad (47)$$

Note that by the uniform convergence (46), we have for any fixed Δ' ,

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\mathbf{P} \in \mathcal{Q}_{\Delta'}} \left| \phi_{\mathbf{P},h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{\mathbf{P},h}(1) \right| = 0$$

Then we use a similar logic from above to conclude

$$\lim_{|\mathcal{W}| \rightarrow \infty} \left| \sup_{\mathbf{P} \in \mathcal{Q}_{\Delta'}} \phi_{\mathbf{P},h}(1) - \sup_{\mathbf{P} \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{\mathbf{P},h}(1) \right| \leq \lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\mathbf{P} \in \mathcal{Q}_{\Delta'}} \left| \phi_{\mathbf{P},h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{\mathbf{P},h}(1) \right| = 0$$

so that

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) = \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \quad (48)$$

The equations (47) and (48) imply that

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) = \sup_{\Delta' \geq \Delta} \lim_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1)$$

Finally we consider $\log \phi_{P,h}(1)$. Since $\|h\|_{\infty} < \infty$, then

$$e^{-\|h\|_{\infty}} \leq e^{-h(Y^{\text{dif}})} \leq e^{\|h\|_{\infty}}$$

hence

$$0 < e^{-\|h\|_{\infty}} \leq \phi_{P,h}(1) = \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] \leq e^{\|h\|_{\infty}}$$

By mean value theorem on $[\phi_{P,h}(1), \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1)] \subseteq [e^{-\|h\|_{\infty}}, e^{\|h\|_{\infty}}]$ for the log function (or $[\lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1), \phi_{P,h}(1)] \subseteq [e^{-\|h\|_{\infty}}, e^{\|h\|_{\infty}}]$),

$$\sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \log \phi_{P,h}(1) - \log \left(\lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right) \right| \leq e^{\|h\|_{\infty}} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right|$$

Since the RHS goes to 0 by results above, then

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \log \phi_{P,h}(1) - \log \left(\lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right) \right| = 0$$

By the continuity of log, we have

$$\log \left(\lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) \right) = \lim_{|\mathcal{W}| \rightarrow \infty} \log \phi_{P,h}(1)$$

thus

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \log \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \log \phi_{P,h}(1) \right| = 0$$

By a similar logic from above,

$$\begin{aligned} & \lim_{|\mathcal{W}| \rightarrow \infty} \left| \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \log \phi_{P,h}(1) - \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \log \phi_{P,h}(1) \right| \\ & \leq \lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \left| \log \phi_{P,h}(1) - \lim_{|\mathcal{W}| \rightarrow \infty} \log \phi_{P,h}(1) \right| \\ & = 0 \end{aligned}$$

Thus we can get the log-version of (47):

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \log \phi_{P,h}(1) = \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \log \phi_{P,h}(1)$$

and the log-version of (48):

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \log \phi_{P,h}(1) = \sup_{P \in \mathcal{Q}_{\Delta'}} \lim_{|\mathcal{W}| \rightarrow \infty} \log \phi_{P,h}(1)$$

Combine we have

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{\Delta' \geq \Delta} \sup_{P \in \mathcal{Q}_{\Delta'}} \log \phi_{P,h}(1) = \sup_{\Delta' \geq \Delta} \lim_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \log \phi_{P,h}(1)$$

Alternatively, we can simply use the fact that $\log$ is continuous and monotone increasing, and interchange with $\lim$ and $\sup$ (this is valid, note all the arguments of $\log$ are positive, and also the limits are positive so we can interchange $\log$ with $\lim$).

Given that the limit exists, the $\limsup$ equals the limit.

Therefore we are done □

Lemma 10 (Lemma 4.3).

For every function h Lipschitz-continuous on $[0, 1]$, we have

$$\bar{R}_{\Delta}(h) \geq - \sup_{\Delta' \geq \Delta} \bar{L}(h, \Delta')$$

Lemma 4.3 implies that maximizing $\bar{R}_{\Delta}(h)$ is related to the minimax optimization: $\inf_h \sup_{\Delta' \geq \Delta} \bar{L}(h, \Delta')$. This minimax formulation is closely related to the original minimax formulation (14), yet it replaces the $|\mathcal{W}|$ -dimensional variable $\mathbf{P}$ with a scalar Δ' after taking asymptotics (Theorem 4.1).

Proof.

Exactly as in the paper. □

Lemma 11 (Lemma C.6).

Let μ_n and μ be probability measures on $\mathbb{R}$ with CDFs given by F_n and F respectively. If μ_n weakly converges to μ and F is continuous, then F_n converges uniformly to F on $\mathbb{R}$.

Lemma 12 (Lemma 4.5).

For any non-increasing and Lipschitz-continuous h on $[0, 1]$, we have

$$\bar{L}(h, \Delta') = \int_0^1 h(r) F_{\text{dif},0}(dr) + \log \int_0^1 e^{-h(r)} F_{\text{dif},\Delta'}(dr)$$

Proof.

Since

$$\bar{L}(h, \Delta') = \limsup_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \left[\mathbb{E}_0 h(Y^{\text{dif}}) + \log \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] \right]$$

and we know $\mathbb{E}_0 h(Y^{\text{dif}})$ does not depends on $\mathbf{P}$, then for fixed $|\mathcal{W}|$ and fixed Δ' ,

$$\begin{aligned} & \sup_{P \in \mathcal{Q}_{\Delta'}} \left[\mathbb{E}_0 h(Y^{\text{dif}}) + \log \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] \right] \\ &= \mathbb{E}_0 h(Y^{\text{dif}}) + \sup_{P \in \mathcal{Q}_{\Delta'}} \log \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] \end{aligned}$$

so

$$\bar{L}(h, \Delta') = \limsup_{|\mathcal{W}| \rightarrow \infty} \left[\mathbb{E}_0 h(Y^{\text{dif}}) + \sup_{P \in \mathcal{Q}_{\Delta'}} \log \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] \right]$$

Since by theorem 4.1,

$$\lim_{|\mathcal{W}| \rightarrow \infty} \mathbb{E}_0 h(Y^{\text{dif}}) = \int_0^1 h(r) F_{\text{dif},0}(dr)$$

then

$$\bar{L}(h, \Delta') = \int_0^1 h(r) F_{\text{dif},0}(dr) + \limsup_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \log \mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right]$$

This is because if $a_n \rightarrow a$, then for $\varepsilon > 0$ and n large enough, $a - \varepsilon \leq a_n \leq a + \varepsilon$. Thus

$a - \varepsilon + b_n \leq a_n + b_n \leq a + \varepsilon + b_n$. By monotonicity of $\limsup$, we have

$a - \varepsilon + \limsup_{n \rightarrow \infty} b_n \leq \limsup_{n \rightarrow \infty} (a_n + b_n) \leq a + \varepsilon + \limsup_{n \rightarrow \infty} b_n$. Therefore

$\limsup_{n \rightarrow \infty} (a_n + b_n) = a + \limsup_{n \rightarrow \infty} b_n$.

Recall integration by parts of Riemann–Stieltjes integral (as noted before, in this paper, it is equivalent to Lebesgue–Stieltjes integral, which is the definition for expectations here):

$$\int_0^1 g(r) d(F(r)) = [g(r)F(r)]_{r=0}^{r=1} - \int_0^1 F(r) d(g(r))$$

Thus

$$\mathbb{E}_{1,P} \left[e^{-h(Y^{\text{dif}})} \right] = \int_0^1 e^{-h(r)} F_{1,P}^{\text{dif}}(dr) = \left[e^{-h(r)} F_{1,P}^{\text{dif}}(r) \right]_{r=0}^{r=1} - \int_0^1 F_{1,P}^{\text{dif}}(r) d(e^{-h(r)})$$

Since h is Lipschitz-continuous, it is continuous and of bounded variation. Hence, by the chain rule for Riemann–Stieltjes integrals, we get

$$\int_0^1 F_{1,P}^{\text{dif}}(r) d(\varphi(h(r))) = \int_0^1 F_{1,P}^{\text{dif}}(r) \varphi'(h(r)) dh(r)$$

Therefore we get (49),

$$\phi_{P,h}(1) = \int_0^1 e^{-h(r)} F_{1,P}^{\text{dif}}(dr) = e^{-h(1)} + \int_0^1 F_{1,P}^{\text{dif}}(r) e^{-h(r)} h(dr) \quad (49)$$

By Theorem 4.1, for any $r \in [0, 1]$ and $P \in \mathcal{Q}_{\Delta'}$,

$$\lim_{|\mathcal{W}| \rightarrow \infty} F_{1,P}^{\text{dif}}(r) = F_{\text{dif},\Delta'}(r)$$

by Portmanteau Theorem, $Y_{|\mathcal{W}|}^{\text{dif}} \xrightarrow{d} Y_{\Delta'}^{\text{dif}}$ and we have weak convergence as well.

By Lemma C.6,

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{r \in [0,1]} |F_{\text{dif},\Delta'}(r) - F_{1,P}^{\text{dif}}(r)| = 0 \quad (50)$$

Therefore by (48), (49) and (50),

$$\lim_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \phi_{P,h}(1) = \lim_{|\mathcal{W}| \rightarrow \infty} \phi_{P,h}(1) = e^{-h(1)} + \int_0^1 F_{\text{dif},\Delta'}(r) e^{-h(r)} h(dr) = \int_0^1 e^{-h(r)} F_{\text{dif},\Delta'}(dr)$$

where the last equation uses integration by parts.

Since $\log$ is continuous and monotone increasing, then we can interchange it with $\lim$ and $\sup$ (this is valid, note all the arguments of $\log$ are positive, and also the limits are positive so we can interchange $\log$ with $\lim$), so we are done. $\square$

Lemma 13 (Lemma 4.6).

For any **non-increasing** and Lipschitz-continuous h on $[0, 1]$,

$$\sup_{\Delta' \geq \Delta} \bar{L}(h, \Delta') = \bar{L}(h, \Delta) = \int_0^1 h(r) F_{\text{dif},0}(dr) + \log \int_0^1 e^{-h(r)} F_{\text{dif},\Delta}(dr)$$

Proof.

Using integration by parts, we first obtain the equation:

$$\int_0^1 e^{-h(r)} F_{\text{dif}, \Delta'}(dr) = e^{-h(1)} - \int_0^1 F_{\text{dif}, \Delta'}(r) e^{-h(r)} (-h)(dr) \quad (1)$$

Observe that we can view $(-h)(dr) = d(-h(r)) = \nu_{-h}(dr)$ as a finite nonnegative Lebesgue–Stieltjes measure because h is non-increasing, so $-h$ is non-decreasing, and hence we have

$$\nu_{-h}((a, b]) := (-h)(b) - (-h)(a) \geq 0$$

Next, take $\Delta'_1 \leq \Delta'_2$, we have

$$F_{\text{dif}, \Delta'_1}(r) \leq F_{\text{dif}, \Delta'_2}(r)$$

Thus

$$\int_0^1 F_{\text{dif}, \Delta'_1}(r) e^{-h(r)} (-h)(dr) \leq \int_0^1 F_{\text{dif}, \Delta'_2}(r) e^{-h(r)} (-h)(dr)$$

As a consequence, we see the integral (1) is non-increasing as Δ' increases.

By Lemma 4.5, this further implies that $\bar{L}(h, \Delta')$ is non-increasing in Δ' . Therefore

$$\sup_{\Delta' \geq \Delta} \bar{L}(h, \Delta') = \bar{L}(h, \Delta)$$

and we are done. □

Lemma 14 (Lemma 4.2).

For any function h that is **non-increasing**, Lipschitz-continuous on $[0, 1]$, there is the lower bound:

$$\bar{R}_\Delta(h) \geq - \left[\int_0^1 h(r) F_{\text{dif}, 0}(dr) + \log \int_0^1 e^{-h(r)} F_{\text{dif}, \Delta}(dr) \right] \quad (26)$$

Proof.

Exactly as in the paper.

We are interested in lower bounding

$$\bar{R}_\Delta(h) = \liminf_{|\mathcal{W}| \rightarrow \infty} R_{\bar{\mathcal{P}}_\Delta}(h)$$

Recall that its definition is given by

$$\bar{R}_\Delta(h) = - \limsup_{|\mathcal{W}| \rightarrow \infty} \inf_{\theta \geq 0} \sup_{P \in \bar{\mathcal{P}}_\Delta} \left[\theta \mathbb{E}_0 h(Y^{\text{dif}}) + \log \mathbb{E}_{1, P} \left[e^{-\theta h(Y^{\text{dif}})} \right] \right] \quad (27)$$

Our lower-bound strategy relies on first connecting $\bar{R}_\Delta(h)$ to the intermediate quantity

$$\bar{L}(h, \Delta') := \limsup_{|\mathcal{W}| \rightarrow \infty} \sup_{P \in \mathcal{Q}_{\Delta'}} \left[\mathbb{E}_0 h(Y^{\text{dif}}) + \log \mathbb{E}_{1, P} \left[e^{-h(Y^{\text{dif}})} \right] \right]$$

where

$$\mathcal{Q}_{\Delta'} := \{ \mathbf{P} : P_{(1)} = 1 - \Delta', \log |\mathcal{W}| \cdot P_{(2)} \leq \varepsilon_{|\mathcal{W}|} \}$$

This connection is given in Lemma 4.3.

Lemma 4.3 reduces the problem of finding a lower bound on $\bar{R}_\Delta(h)$ to evaluating $\sup_{\Delta' \geq \Delta} \bar{L}(h, \Delta')$. To do so, we first derive an explicit expression for $\bar{L}(h, \Delta')$, presented in Lemma 4.5.

Finally, with Lemma 4.5 we are able to derive an explicit expression of the supremum $\sup_{\Delta' \geq \Delta} \bar{L}(h, \Delta')$, presented in Lemma 4.6.

By Lemma 4.3 and Lemma 4.6, we obtain

$$\bar{R}_\Delta(h) \geq - \sup_{\Delta' \geq \Delta} \bar{L}(h, \Delta') = - \int_0^1 h(r) F_{\text{dif},0}(dr) - \log \int_0^1 e^{-h(r)} F_{\text{dif},\Delta}(dr)$$

This yields the lower bound as desired. □

Theorem 15 (Theorem 4.2).

Fix $\Delta \in (0, 1]$. Let $h_{\text{dif},\Delta}^* : [0, 1] \rightarrow \mathbb{R}$ denote

$$h_{\text{dif},\Delta}^*(r) = \log \frac{f_{\text{dif},\Delta}(r)}{f_{\text{dif},0}(r)}, \quad \text{where } f_{\text{dif},\Delta}(r) = \frac{2}{1-\Delta} \max \left\{ 1 - \frac{r}{1-\Delta}, 0 \right\}$$

Then,

$$\lim_{M \rightarrow \infty} \bar{R}_\Delta \left([h_{\text{dif},\Delta}^*]_{[-M,M]} \right) = \infty$$

where $[x]_{[-M,M]} := \min\{\max\{x, -M\}, M\}$ denotes the clipping operator that restricts x to the interval $[-M, M]$.

Proof.

Fix $\Delta \in (0, 1]$. To prove Theorem 4.2, our goal is to establish the limit:

$$\lim_{M \rightarrow \infty} \bar{R}_\Delta \left([h_{\text{dif},\Delta}^*]_{[-M,M]} \right) = \infty$$

Our strategy involves establishing an effective lower bound for $\bar{R}_\Delta(h)$, as detailed in Lemma 4.2. This involves introducing a function $F_{\text{dif},\Delta}$ for each $\Delta \in [0, 1]$, defined as:

$$F_{\text{dif},\Delta}(r) = 1 - \max \left\{ 1 - \frac{r}{1-\Delta}, 0 \right\}^2 \quad \text{for any } r \in [0, 1].$$

By Theorem 4.1, this corresponds to the limiting CDF of Y^{dif} given $\mathbf{P}_t$, assuming that $\mathbf{P}_t \in \bar{\mathcal{P}}_\Delta$.

Now we finish the proof of Theorem 4.2. Let $G(h)$ denote the RHS of inequality (26) in Lemma 4.2. Also, let $\mu_{\text{dif},\Delta}$ denote the probability measure associated with the CDF $F_{\text{dif},\Delta}$.

By the Extended Donsker–Varadhan representation, we deduce:

$$\begin{aligned} \sup_h G(h) &= \sup_h \left[- \int_0^1 h(r) F_{\text{dif},0}(dr) - \log \int_0^1 e^{-h(r)} F_{\text{dif},\Delta}(dr) \right] \\ &= D_{\text{KL}}(\mu_{\text{dif},0}, \mu_{\text{dif},\Delta}) \\ &= \infty \end{aligned}$$

where the last identity holds because: The alternative limiting distribution is supported on $[0, 1 - \Delta]$, so

$$\mu_{\text{dif},\Delta}((1 - \Delta, 1]) = 0$$

But the null limiting CDF is $F_{\text{dif},0}(r) = 1 - (1 - r)^2$, thus

$$\begin{aligned} \mu_{\text{dif},0}((1 - \Delta, 1]) &= 1 - F_{\text{dif},0}(1 - \Delta) \\ &= 1 - [1 - \Delta^2] \\ &= \Delta^2 > 0 \end{aligned}$$

Therefore

$$\mu_{\text{dif},0} \not\ll \mu_{\text{dif},\Delta}$$

and

$$D_{\text{KL}}(\mu_{\text{dif},0} \| \mu_{\text{dif},\Delta}) = \infty$$

Define the extended-real-valued function

$$h_{\text{dif},\Delta}^*(r) = \begin{cases} \log \frac{1-\Delta-r}{(1-\Delta)^2(1-r)}, & 0 \leq r < 1 - \Delta, \\ -\infty, & 1 - \Delta \leq r \leq 1 \end{cases}$$

For $\mu_{\text{dif},0}$ -almost every r ,

$$h_{\text{dif},\Delta}^* = \log \frac{d\mu_{\text{dif},\Delta}}{d\mu_{\text{dif},0}} = \log \frac{f_{\text{dif},\Delta}}{f_{\text{dif},0}}$$

Denote $S = [0, 1 - \Delta)$ and $A = [1 - \Delta, 1]$.

Consider

$$h_M = [h_{\text{dif},\Delta}^*]_{[-M,M]}$$

For every finite M , the function h_M is real-valued, non-increasing, and Lipschitz-continuous. Therefore, Lemma 4.2 gives

$$\overline{R}_\Delta(h_M) \geq G(h_M)$$

Since $h_{\text{dif},\Delta}^* = -\infty$ on A , we have $h_M = -M$ on A by definition of clipping operator.

Moreover, recall

$$\mu_{\text{dif},0}(A) = \mu_{\text{dif},0}((1 - \Delta, 1]) = \Delta^2$$

For $r \in S = [0, 1 - \Delta)$,

$$h_{\text{dif},\Delta}^*(r) = \log \left[\frac{1 - \Delta - r}{(1 - \Delta)^2(1 - r)} \right]$$

Since

$$\frac{d}{dr} h_{\text{dif},\Delta}^*(r) = -\frac{\Delta}{(1 - \Delta - r)(1 - r)} < 0$$

then

$$h_{\text{dif},\Delta}^*(r) \leq h_{\text{dif},\Delta}^*(0) = \log \frac{1}{1 - \Delta}$$

Choose $M > 0$ large enough such that $M \geq \log \frac{1}{1 - \Delta}$ and $-M \leq 0 \leq \log \frac{1}{1 - \Delta}$. then we have $h_{\text{dif},\Delta}^*(r) \leq M$, hence

$$h_M(r) = \max \{h_{\text{dif},\Delta}^*(r), -M\}$$

Therefore

$$h_M(r) \geq h_{\text{dif},\Delta}^*(r)$$

The measure $\mu_{\text{dif},\Delta}$ has density $f_{\text{dif},\Delta}$ with respect to Lebesgue measure:

$$\mu_{\text{dif},\Delta}(dr) = f_{\text{dif},\Delta}(r) dr$$

For $\mu_{\text{dif},0}$ -almost every $r \in S$,

$$e^{-h_{\text{dif},\Delta}^*(r)} = \exp \left(-\log \frac{d\mu_{\text{dif},\Delta}}{d\mu_{\text{dif},0}}(r) \right) = \left[\left(\frac{d\mu_{\text{dif},\Delta}}{d\mu_{\text{dif},0}} \right)(r) \right]^{-1} = \left[\frac{f_{\text{dif},\Delta}(r)}{f_{\text{dif},0}(r)} \right]^{-1} = \frac{f_{\text{dif},0}(r)}{f_{\text{dif},\Delta}(r)}$$

Since $\mu_{\text{dif},\Delta} \ll \mu_{\text{dif},0}$, the preceding identity also holds for $\mu_{\text{dif},\Delta}$ -almost every $r \in S$.

Therefore

$$\begin{aligned}
\int_S e^{-h_M(r)} \mu_{\text{dif},\Delta}(dr) &\leq \int_S e^{-h_{\text{dif},\Delta}^*(r)} \mu_{\text{dif},\Delta}(dr) \\
&= \int_S \frac{f_{\text{dif},0}(r)}{f_{\text{dif},\Delta}(r)} \mu_{\text{dif},\Delta}(dr) \\
&= \int_0^{1-\Delta} \frac{f_{\text{dif},0}(r)}{f_{\text{dif},\Delta}(r)} f_{\text{dif},\Delta}(r) dr \\
&= \int_0^{1-\Delta} f_{\text{dif},0}(r) dr \\
&= \int_0^{1-\Delta} 2(1-r) dr \\
&= [2r - r^2]_0^{1-\Delta} \\
&= 2(1-\Delta) - (1-\Delta)^2 \\
&= 1 - \Delta^2 \\
&\leq 1
\end{aligned}$$

Therefore, the negative logarithmic term in $G(h_M)$ is nonnegative. So when finding the lower bound for $G(h_M)$, we can omit this term.

On A ,

$$h_M(r) = -M$$

and

$$\mu_{\text{dif},0}(A) = \Delta^2$$

Thus

$$\begin{aligned}
-\int_A h_M(r) \mu_{\text{dif},0}(dr) &= -\int_A (-M) \mu_{\text{dif},0}(dr) \\
&= M \mu_{\text{dif},0}(A) \\
&= M \Delta^2
\end{aligned}$$

On S , recall we had

$$h_M(r) = \max \{h_{\text{dif},\Delta}^*(r), -M\}$$

where M is chosen large enough such that $M \geq \log \frac{1}{1-\Delta}$ and $-M \leq 0 \leq \log \frac{1}{1-\Delta}$. Also recall we had $h_{\text{dif},\Delta}^*(r) \leq \log \frac{1}{1-\Delta}$.

Therefore the maximum

$$h_M(r) \leq \log \frac{1}{1-\Delta}$$

for $r \in S$. Thus

$$\begin{aligned}
\int_S h_M(r) \mu_{\text{dif},0}(dr) &\leq \log \frac{1}{1-\Delta} \mu_{\text{dif},0}(S) \\
&\leq \log \frac{1}{1-\Delta}
\end{aligned}$$

Therefore

$$\begin{aligned}
G(h_M) &= \underbrace{-\int_S h_M d\mu_{\text{dif},0}}_{\geq -\log \frac{1}{1-\Delta}} + \underbrace{\left(-\int_A h_M d\mu_{\text{dif},0}\right)}_{= M \Delta^2} + \underbrace{\left(-\log \int_S e^{-h_M} d\mu_{\text{dif},\Delta}\right)}_{\geq 0} \geq M \Delta^2 - \log \frac{1}{1-\Delta}
\end{aligned}$$

Since $\Delta > 0$, as $M \rightarrow \infty$,

$$G(h_M) \rightarrow \infty$$

Hence

$$\bar{R}_\Delta(h_M) \geq G(h_M) \rightarrow \infty$$

which proves

$$\lim_{M \rightarrow \infty} \overline{R}_\Delta \left([h_{\text{dif}, \Delta}^*]_{[-M, M]} \right) = \infty$$

And we are finally done. □

Theorem 16 (Extended Donsker–Varadhan).

Let P and Q be probability measures on a measurable space. Their Kullback–Leibler divergence is defined by

$$D_{\text{KL}}(P \| Q) = \begin{cases} \int_{\Omega} \log \left(\frac{dP}{dQ} \right) dP, & P \ll Q \\ +\infty, & P \not\ll Q \end{cases}$$

The Extended Donsker–Varadhan variational representation states that

$$D_{\text{KL}}(P \| Q) = \sup_{g \in \mathcal{B}_b(\Omega)} \left\{ \int_{\Omega} g dP - \log \int_{\Omega} e^g dQ \right\}$$

where $\mathcal{B}_b(\Omega)$ denotes the set of all bounded measurable real-valued functions on the underlying space.

The equality is understood in the extended real numbers. Therefore, it remains valid when either side is infinite.

Proof.

We show the infinite case.

Suppose that P is not absolutely continuous with respect to Q . Then there exists a measurable set A such that $P(A) > 0$ and $Q(A) = 0$.

For every positive integer M , define

$$g_M = M \mathbf{1}_A$$

Each g_M is bounded and measurable. Its Donsker–Varadhan objective is

$$\begin{aligned} \int_{\Omega} g_M dP - \log \int_{\Omega} e^{g_M} dQ &= MP(A) - \log (Q(A^c) + e^M Q(A)) \\ &= MP(A) - \log 1 \\ &= MP(A) \end{aligned}$$

Because $P(A)$ is strictly positive, $MP(A) \rightarrow \infty$ as $M \rightarrow \infty$.

Consequently,

$$\sup_{g \in \mathcal{B}_b(\Omega)} \left\{ \int_{\Omega} g dP - \log \int_{\Omega} e^g dQ \right\} = \infty = D_{\text{KL}}(P \| Q)$$

□